

Measures of Disease Occurrence

Question: 1 of 5 [Qid : 1]

A new combined chemotherapy and immunotherapy regimen has been shown to significantly prolong survival in patients with metastatic melanoma. If widely implemented, which of the following changes in disease occurrence measures would you most expect?

- ☐ A) Incidence increases, prevalence decreases
- ☐ B) Incidence decreases, prevalence decreases
- ☐ C) Incidence increases, prevalence increases
- ☐ D) Incidence does not change, prevalence increases
- ☐ E) Incidence does not change, prevalence does not change

Question: 2 of 5 [Qid : 2]

The incidence of diabetes mellitus in a population with very little migration has remained stable over the past 40 years (55 cases per 1000 people per year). At the same time, prevalence of the disease increased threefold over the same period. Which of the following is the best explanation for the changes in diabetes occurrence measures in the population?

- ☐ A) Increased diagnostic accuracy
- ☐ B) Poor event ascertainment
- ☐ C) Improved quality of care
- ☐ D) Increased overall morbidity
- ☐ E) Loss at follow-up

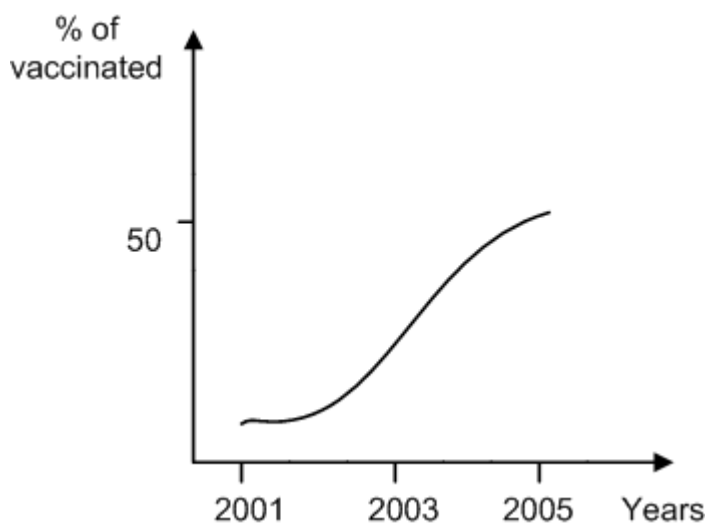
Question: 3 of 5 [Qid : 3]

In a survey of 10,000 IV drug abusers in town A, 1,000 turn out to be infected with hepatitis C and 500 infected with hepatitis B. During two years of follow-up, 200 patients with hepatitis C infection and 100 patients with hepatitis B infection die. Also during follow-up, 200 IV drug abusers acquire hepatitis C and 50 acquire hepatitis B. Which of the following is the best estimate of the annual incidence of hepatitis C infection in IV drug abusers in town A?

- ☐ A) 1,000/10,000
- ☐ B) 1,100/10,000
- ☐ C) 100/10,000
- ☐ D) 100/9,000
- ☐ E) 100/9,800

Question: 4 of 5 [Qid : 4]

The following graph represents the vaccination rate dynamics for hepatitis B in IV drug abusers in town A.



Which of the following hepatitis D statistics is most likely to be affected by the reported data?

- ☐ A) Hospitalization rate
- ☐ B) Case fatality rate
- ☐ C) Median survival
- ☐ D) Incidence
- ☐ E) Cure rate

Question: 5 of 5 [Qid : 5]

In a city having a population of 1,000,000 there are 300,000 women of childbearing age. The following statistics are reported for the city in the year 2000:

Fetal deaths: 200

Live births: 5,000

Maternal deaths: 70

Which of the following is the best estimate of the maternal mortality rate in the city in the year 2000?

- ☐ A) 70/1,000,000
- ☐ B) 70/300,000
- ☐ C) 70/5,000
- ☐ D) 70/5,200

Correct Answers: 1) D 2) C 3) D 4) D 5) C

Explanation :

Two basic measures of disease occurrence in a population are incidence and prevalence. Although simple in definition, they are frequently confused with each other. Moreover, many USMLE questions are based on simple understanding of these basic measures.

Incidence measures *new cases* that develop in a population over a certain *period* of time. It is important to define the period of time during which the number of new cases is counted (e.g., weekly incidence vs annual incidence). Incidence does not take into account the number of cases that already existed in the population before the counting period began. It is also important to include in the denominator only the population at risk of acquiring the disease. For example, in Question #3, IV drug abusers diagnosed with hepatitis C infection before the follow-up period began should be excluded from the denominator because they already have the disease and thus are no longer 'at risk' (10,000 - 1,000). The best estimate of the annual incidence would be 100/9,000 because 200 new hepatitis C cases have been diagnosed over the TWO year follow-up period.

Figure 1 and Figure 2 demonstrate the difference between incidence and prevalence diagrammatically. Figure 1 contains two arrows demarcating the one year time frame during which the number of new cases is to be measured. You can see that three new cases have been identified during this period, making the annual incidence 3 cases per year.

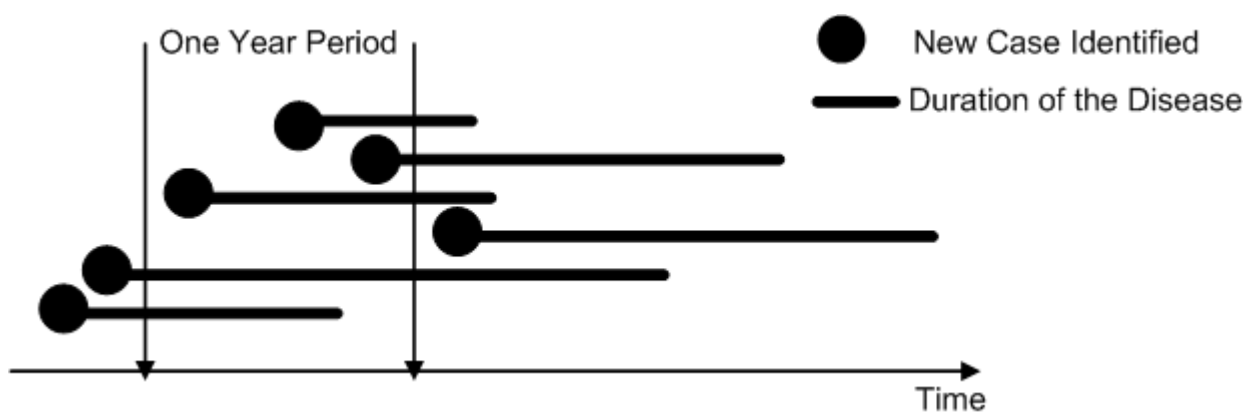


Fig.1. Three new cases have been identified during the one year period, making incidence 3 cases per year.

Prevalence of a disease is a measure of the *total number* of cases (new and old) measured at a particular *point* in time. You can conceptualize it as a 'snapshot' of the number of diseased individuals at a given point of time (**Figure 2**).

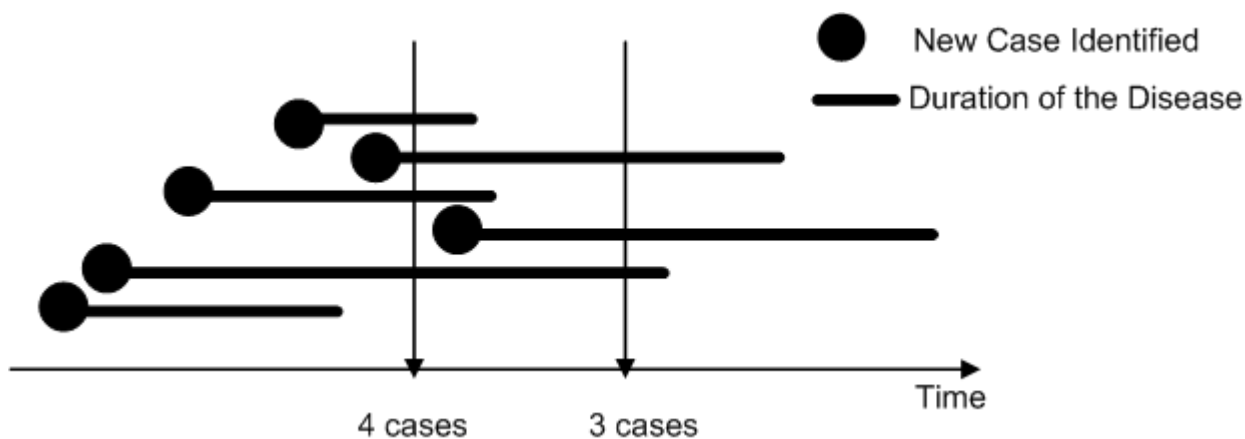


Fig.2. Prevalence of a disease is a 'snapshot' of the total number of diseased individuals at a given point of time.

You can also tell from **Figures 1 and 2** that prevalence and incidence are related to each other. Prevalence is a function of both the incidence and duration of the disease. Diseases that have a short duration due to high mortality (e.g., aggressive cancer) or quick convalescence (e.g., the flu) tend to have low prevalence, even if incidence is high. At the same time, chronic diseases (e.g., hypertension and diabetes) tend to have high prevalence, even if incidence is low.

Chronic disease treatments that prolong patient survival increase the prevalence of disease due to accumulation of cases over time; incidence is not affected by such treatments because it measures only new cases as they arise. Increasing prevalence of a chronic disease despite stable incidence is usually related to improved quality of care and resultant decrease in mortality. Improved diagnostic accuracy for a chronic disease leads to both increased incidence (more cases are identified) and prevalence. Primary prevention (e.g., hepatitis vaccination) decreases incidence of the disease, and also eventually decreases prevalence as patients with disease that predates primary prevention die or attain cure.

Some specific measures of disease occurrence are explained below:

- Crude mortality rate: Calculated by dividing the number of deaths by the total population size.
- Cause-specific mortality rate: Calculated by dividing the number of deaths from a particular disease by the total population size.
- Case-fatality rate: Calculated by dividing the number of deaths from a specific disease by the number of people affected by the disease.
- Standardized mortality ratio (SMR): Calculated by dividing the observed number of deaths by the expected number of deaths. This measure is used sometimes in occupational epidemiology. SMR of 2.0 indicates that the observed mortality in a particular group is twice as high as that in the general population.
- Attack rate: An incidence measure typically used in infectious disease epidemiology. It is calculated by dividing the number of patients with disease by the total population at risk. For example, attack rate can be calculated for gastroenteritis among people who ate contaminated food.
- Maternal mortality rate: Calculated by dividing the number of maternal deaths by the number of live births (see Question #5).
- Crude birth rate: Defined as the number of live births divided by the total population size.

Odds Ratio and Relative Risk

Question: 1 of 3 [Qid : 6]

An observational study in diabetics assesses the role of an increased plasma fibrinogen level on the risk of cardiac events. 130 diabetic patients are followed for 5 years to assess for the development of acute coronary syndrome. In a group of 60 patients with a normal baseline plasma fibrinogen level, 20 develop acute coronary syndrome and 40 do not. In a group of 70 patients with a high baseline plasma fibrinogen level, 40 develop acute coronary syndrome and 30 do not. Which of the following is the best estimate of relative risk in patients with a high baseline plasma fibrinogen level compared to patients with a normal baseline plasma fibrinogen level?

- ☐ A) $(40/30)/(20/40)$
- ☐ B) $(40 \times 40)/(20 \times 30)$
- ☐ C) $(40 \times 70)/(20 \times 60)$
- ☐ D) $(40/70)/(20/60)$
- ☐ E) $(40/60)/(20/70)$

Question: 2 of 3 [Qid : 7]

A study is performed in which mothers of babies born with neural tube defects are questioned about their acetaminophen consumption during the first trimester of pregnancy. At the same time, mothers of babies born without neural tube defect are also questioned about their consumption of acetaminophen during the first trimester. Which of the following measures of association is most likely to be reported by investigators?

- ☐ A) Prevalence ratio
- ☐ B) Median survival
- ☐ C) Relative risk
- ☐ D) Odds ratio
- ☐ E) Hazard ratio

Question: 3 of 3 [Qid : 8]

At a specific hospital, patients diagnosed with pancreatic carcinoma are asked about their current smoking status. At the same hospital, patients without pancreatic carcinoma are also asked about their current smoking status. The following table is constructed.

	Smokers	Non-smokers	Total
Pancreatic cancer	50	40	90
No pancreatic cancer	60	80	140
Total	110	120	230

What is the odds ratio that a patient diagnosed with pancreatic cancer is a current smoker compared to a patient without pancreatic cancer?

- ☐ A) $(50/90)/(60/140)$
- ☐ B) $(50/40)/(60/80)$

- ☐ C) (50/110)/(40/120)
- ☐ D) (50/60)/(40/80)
- ☐ E) (90/230)/(140/230)

Correct Answers: 1) D 2) D 3) B

Explanation :

- Two basic measures of association that you should be familiar with are relative risk (or risk ratio) and odds ratio. You should be able to both calculate and interpret them.
- Risk refers to the probability of an event occurring over a certain period of time. Therefore, it typically implies a prospective study design. In Question #1, diabetic patients are followed over 5 years to assess for the development of acute coronary syndrome; that means it is possible to calculate and report 5-year risk of acute coronary events in these patients. Moreover, we can compare the 5-year risk of developing acute coronary syndrome in patients with a high baseline fibrinogen level (exposure group) to the patients with a normal baseline fibrinogen level (non-exposure group).
- In case-control studies (like the one described in Question #2) patients are not followed over time to determine their outcome. Rather, the outcome (babies with neural tube defect) is known from the start of the study. Therefore it is impossible to calculate risk in such studies, but it is possible to inquire about past exposures. In case-control studies, we calculate the odds of exposure (the chance of being exposed to a particular factor) in case patients (those with disease) and compare it with the odds of exposure in control patients (those without disease). For example, in Question #2 we can calculate the odds of acetaminophen use in mothers having babies with a neural tube defect (cases) to mother having normal babies (controls).
- In summary, relative risk compares the probability of developing an outcome between two groups over a certain period of time. It implies a prospective study design because the patients are followed over time to see whether or not they develop an outcome. Odds ratio compares the chance of exposure to a particular risk factor in cases and controls. Since risk can not be calculated directly in case-control studies (because they are not prospective), odds ratio is the measure of association used for this study design. Relative risk answers the question: within certain period of time, how many times are exposed people more likely to develop a particular event compared to unexposed people? Odds ratio answers the questions: how many times are diseased people more likely to be exposed to a particular factor compared to non-diseased people? Both relative risk and odds ratio are measured on a scale from 0 to infinity. The value of 1.0 indicates no difference between the two groups being compared. Odds ratio approximates relative risk when the disease under study is rare (so called 'rare disease assumption').
- Calculating measures of association from the data presented in clinical cases requires several consecutive steps. The first step is to identify exposure and outcome. In Question #1, baseline plasma fibrinogen level is the exposure of interest and acute coronary event is the outcome (disease) of interest. The second step is to group study subjects into the following categories: exposed diseased; exposed non-diseased; unexposed diseased; and unexposed non-diseased. In Question #1, the groups would contain 40, 30, 20 and 40 patients, respectively. The third step is to construct a 2*2 table based on the grouping described above (see the table).

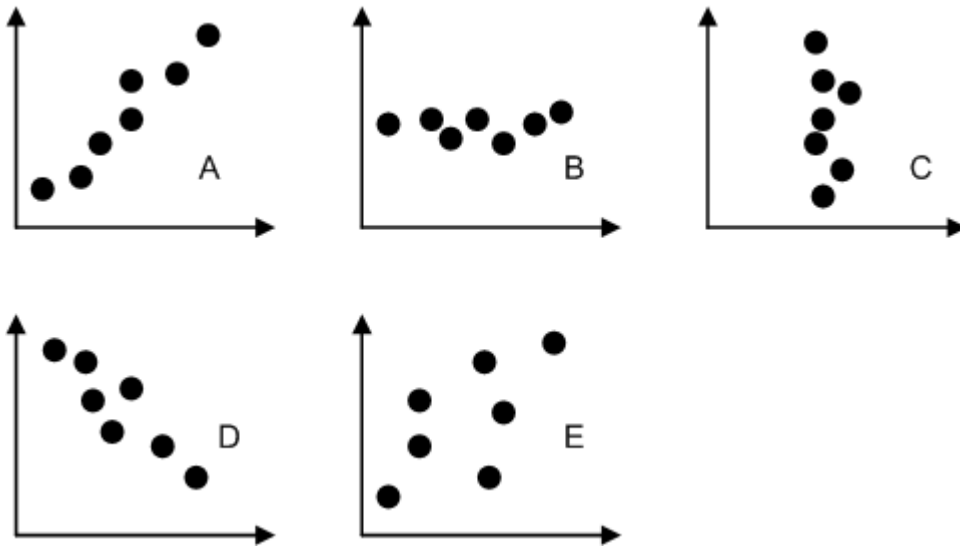
	Exposed	Unexposed	Total
Diseased	40 (a)	20 (c)	60
Non-diseased	30 (b)	40 (d)	70
Total	70	60	130

- The final step is the actual calculation.
- To determine relative risk you compare the risk of disease in exposed subjects ($a/(a+b)$) with the risk of disease in unexposed subjects ($c/(c+d)$). In Question #1, the relative risk is therefore: $(40/70)/(20/60)$.
- To determine exposure odds ratio you compare the odds of exposure in diseased subjects (a/c) with the odds of exposure in non-diseased subjects (b/d). In Question #3, the odds of being a smoker for a patient with pancreatic cancer are 50/40, whereas the odds of being a smoker for a patient without pancreatic cancer are 60/80. Therefore, the odds ratio is best expressed as: $(50/40)/(60/80) = 1.7$.
- The odds ratio equation can also be rearranged in the following manner with the same final result: odds ratio = ad/bc . In Question #3 it would be calculated as: $(50*80)/(40*60) = 1.7$.

Correlation

Question: 1 of 3 [Qid : 9]

Which of the following graphs most closely corresponds to a correlation coefficient of + 1.0?



- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D
- ☐ E) E

Question: 2 of 3 [Qid : 10]

A group of investigators describes a linear association between calcium content of the aortic valve cusps as measured *in vivo* and the diameter of the aortic opening. They report a correlation coefficient of -0.45 and a p value of 0.001. Which of the following is the best interpretation of the results reported by the investigators?

- ☐ A) Alpha-error level is set too low
- ☐ B) Sample size is too low for drawing definite conclusions
- ☐ C) Calcium deposition causes narrowing of the aortic valve opening
- ☐ D) As calcium content of the cusps increases the aortic valve diameter decreases
- ☐ E) As aortic valve diameter decreases the calcium content of the cusps decreases

Question: 3 of 3 [Qid : 11]

A study is conducted to assess the relationship between plasma homocysteine level and folic acid intake. The investigators demonstrate that the plasma homocysteine level is inversely related to folic acid intake, and the correlation coefficient is -0.8 ($p < 0.01$). According to the information provided, how much of the variability in plasma homocysteine levels is explained by folic acid intake?

- ☐ A) > 0.99

- ☐ B) 0.80
- ☐ C) 0.64
- ☐ D) 0.55
- ☐ E) < 0.01

Correct Answers: 1) A 2) D 3) C

Explanation :

Scatter plots, as demonstrated in Question #1, are useful for crude analysis of data. They can be used to demonstrate whether any type of association (i.e., linear, non-linear) exists between two continuous variables. Examples of continuous variables for which an association can be demonstrated are: arterial blood pressure and dietary salt consumption; blood glucose level and blood C-peptide level; etc. If a linear association is present, the correlation coefficient can be calculated to provide a numerical description of the linear association.

The correlation coefficient ranges from -1 to +1 and describes two important characteristics of an association: the strength and polarity. For example, in Question #1, graph A describes a strong positive association (as the value of one variable increases the value of the other variable also increases) whereas graph D describes a strong negative association (as the value of one variable increases the value of the other variable decreases). Graph E describes a weaker positive association compared to graph A; you should expect a correlation coefficient around +0.5. Graphs B and C demonstrate no correlation because the value of one variable stays the same over the range of values of the other variable.

You can also calculate the coefficient of determination by squaring the correlation coefficient. The coefficient of determination expresses the percentage of the variability in the outcome factor that is explained by the predictor factor. In Question #3, 0.64 (64%) of variability in plasma homocysteine level is explained by folic acid intake.

It is important to note that a correlation coefficient describes a linear association but it does not necessarily imply causation. This explains why answer choice D is superior to choice C in Question #2.

Attributable Risk

Question: 1 of 4 [Qid : 12]

In a small observational study, 100 industrial workers are followed for one year to assess for the development of respiratory symptoms (defined as productive cough lasting at least one week). 30 of 60 smokers experience respiratory symptoms over the year versus 10 of 40 non-smokers. Which of the following is the best estimate of the attributable risk of respiratory disease in smokers?

- ☐ A) 0.75
- ☐ B) 0.50
- ☐ C) 0.25
- ☐ D) 0.30
- ☐ E) 0.10

Question: 2 of 4 [Qid : 13]

In a small observational study, 100 industrial workers are followed for one year to assess for the development of respiratory symptoms (defined as productive cough lasting at least one week). 30 of 60 smokers experience respiratory symptoms over the year versus 10 of 40 non-smokers. What percentage of respiratory disease experienced by smokers is attributed to smoking?

- ☐ A) 90%
- ☐ B) 75%
- ☐ C) 50%
- ☐ D) 25%
- ☐ E) 10%

Question: 3 of 4 [Qid : 14]

In a small observational study, 100 industrial workers are followed for one year to assess for the development of respiratory symptoms (defined as productive cough lasting at least one week). 30 of 60 smokers experience respiratory symptoms over the year versus 10 of 40 non-smokers. What percentage of respiratory disease experienced by all study subjects is attributed to smoking?

- ☐ A) 75%
- ☐ B) 50%
- ☐ C) 25%
- ☐ D) 20%
- ☐ E) 10%

Question: 4 of 4 [Qid : 15]

A new chemotherapy regimen used in patients with ovarian carcinoma is tested in a small clinical trial. Out of 50 patients treated with the new regimen, 25 survive 5 years without relapse. Out of 100 patients treated with the conventional regimen, 25 survive 5 years without relapse. How many patients need to be treated with the new regimen as opposed to the conventional regimen in order for one more patient to survive 5

years without relapse?

- ☐ A) 2
- ☐ B) 4
- ☐ C) 6
- ☐ D) 8
- ☐ E) 10

Correct Answers: 1) C 2) C 3) D 4) B

Explanation :

Several important topics related to measures of association and impact are covered in this section.

The first topic is known as 'attributable risk' or 'risk difference'. It is a measure of the excess incidence of a disease due to a particular factor (exposure). In Question #1, the one-year incidence of respiratory disease in smokers is $30/60 = 0.5$ whereas in non-smokers it is $10/40 = 0.25$. The difference between these incidences ($0.5 - 0.25 = 0.25$) describes the attributable risk. Based on the calculation, we can assume that 25 out of 100 cases of respiratory disease in smokers are attributable to smoking.

A related measure known as 'attributable risk percent' describes the contribution of a given exposure to the incidence of a disease in *relative* terms. Attributable risk percent is calculated by dividing the attributable risk by the incidence of the disease in the exposed population (i.e. smokers). In Question #2 we calculate attributable risk percent as follows: $(30/60 - 10/40)/(30/60) = 0.25/0.5 = 0.5$ (50%). Based on the calculation, we can conclude that 50% of the yearly respiratory disease in smokers is attributable to smoking.

Another measure called population attributable risk percent describes the impact of exposure on the entire study population (in our case, both smokers and non-smokers). To determine population attributable risk percent, first calculate the incidence of the disease in the study population as a whole. In the above study population, there are 30 smokers and 10 non-smokers who develop respiratory disease out of a total of 100 workers. Therefore, the overall incidence of respiratory disease in the study population is $40/100$. Next, calculate the difference in risk of developing respiratory disease between smokers and the study population as a whole ($30/60 - 40/100 = 0.5 - 0.4 = 0.1$) and divide this value by the incidence of respiratory disease in smokers ($0.1/0.5 = 0.2$). Based on the calculation, we conclude that 20% of the yearly respiratory disease in the study population is attributable to smoking. (Note: if one obtains the relative risk, attributable risk percent can be calculated as follows: attributable risk percent = $(RR - 1)/RR$).

In clinical trials, an important concept related to absolute risk reduction is 'number needed to treat' (NNT). It is actually the reciprocal of absolute risk reduction. It answers the following question: how many patients should I treat with the drug (or regimen) of interest to save/extend one life? In Question #4 the death rate in patients placed on the new treatment regimen is $25/50 = 0.5$ over 5 years, whereas in patients kept on the conventional chemotherapy regimen the mortality rate is $75/100 = 0.75$. The absolute risk difference between the two groups is $0.75 - 0.5 = 0.25$. The reciprocal of the absolute risk difference ($1/0.25 = 4$) reveals the NNT. Based on this result, we can conclude that we need to treat 4 patients with the new

regimen as opposed to the conventional regimen in order for one more patient to survive 5 years without relapse.

Null Hypothesis and P value

Question: 1 of 2 [Qid : 16]

A group of investigators conducts a study to evaluate the association between serum homocysteine level and the risk of myocardial infarction. They conclude that a high baseline plasma homocysteine level is associated with an increased risk of myocardial infarction and report a risk ratio (RR) of 1.08 and a p value of 0.01. Which of the following is the most accurate statement about the results of the study?

- ☐ A) There is an 8% chance that increased homocysteine levels cause myocardial infarction
- ☐ B) There is a 1% probability that there is no association
- ☐ C) The 95% confidence interval for the RR includes 1.0
- ☐ D) The study has insufficient power to reach a definite conclusion
- ☐ E) There is a 10% probability that the association is underestimated

Question: 2 of 2 [Qid : 17]

High plasma C-reactive protein (CRP) level is believed to be associated with increased risk of acute coronary syndromes. A group of investigators is planning a study that would evaluate that association, taking into account a set of potential confounders. Which of the following is the best statement of null hypothesis for the study?

- ☐ A) High plasma CRP level carries increased risk of acute coronary syndromes
- ☐ B) High plasma CRP level is related to the occurrence of acute coronary syndromes
- ☐ C) High plasma CRP level has no association with acute coronary syndrome
- ☐ D) Acute coronary syndrome can be predicted by high plasma CRP
- ☐ E) High plasma CRP level can cause acute coronary syndromes

Correct Answers: 1) B 2) C

Explanation :

A clear expression of the null hypothesis (H_0) is essential before conducting any study. The null hypothesis typically states that there is *no association* between the exposure of interest and the outcome. For example, if a study is conducted to assess the risk of myocardial infarction in patients taking aspirin versus in patients not taking aspirin, the null hypothesis would be: there is no association between aspirin treatment and the risk of myocardial infarction. Unlike the null hypothesis that denies any association, the alternative hypothesis (H_a) states that the exposure is in some way related to the outcome. The alternate hypothesis can specify whether the exposure increases or decreases the likelihood of the outcome (one-way hypothesis) or it can state that there is an association without specifying its direction (two-way hypothesis).

After data is collected, statistical analysis is then performed. Based on the results of statistical analysis we either accept or reject the null hypothesis. For the purpose of the USMLE board exams, when asked to

interpret the null hypothesis you will typically be provided with the p value and/or confidence interval. P value represents the probability that the null hypothesis is true. For example, if the investigators in the aspirin study report a p value of 0.01, this means that there is a 1% probability that there is no association between aspirin and the risk of myocardial infarction.

To accept or reject the null hypothesis compare the p value to the pre-set alpha level (see the description of alpha error in section 19, Statistical Power). Most investigators believe that an alpha level of 0.05 (or 5%) is an acceptable threshold for statistical significance (assume an alpha level of 0.05 unless otherwise stated). In other words, if the p value is less than 0.05, then there is < 5% probability that the null hypothesis holds true, and we therefore reject the null hypothesis and accept the appropriate alternative hypothesis. Remember, however, that even a very low p value indicates that there is some probability that the null hypothesis is true.

The relationship between p value and confidence interval is described later.

Confidence Interval

Question: 1 of 3 [Qid : 18]

Two studies are conducted to assess the risk of developing asymptomatic liver mass in women taking oral contraceptive pills (OCP). Study A reports a relative risk of 1.6 (95% confidence interval 1.1-2.8) in women taking OCP compared to women not taking OCP over a five-year follow-up period. Study B reports a relative risk of 1.5 (95% confidence interval 0.8-3.5) in women taking OCP compared to women not taking OCP over a five-year follow-up period. Which of the following statements about the two studies is most accurate?

- ☐ A) Study A overestimates the risk
- ☐ B) The result in study B proves no causality
- ☐ C) The result in study A is not accurate
- ☐ D) The sample size in study B is small
- ☐ E) The p value in study B is less than 0.05

Question: 2 of 3 [Qid : 19]

A ten-year prospective study is conducted to assess the effect of regular supplementary folic acid consumption on the risk of developing Alzheimer's dementia. The investigators report a relative risk of 0.77 (95% confidence interval 0.59-0.98) in those who consume folic acid supplements compared to those who do not. Which of the following p values most likely corresponds to the results reported by the investigators?

- ☐ A) 0.03
- ☐ B) 0.05
- ☐ C) 0.07
- ☐ D) 0.09
- ☐ E) 0.15

Question: 3 of 3 [Qid : 20]

A double-blind clinical study is conducted in patients with chronic heart failure, class II and III, treated with an ACE inhibitor and a loop diuretic. The patients are divided into two groups: one group receives metoprolol and the other group receives placebo. The following relative risk values are reported for the metoprolol group compared to the placebo group:

	Relative Risk	Confidence Interval
All-cause mortality	0.89	0.79 – 1.01
Myocardial infarction	0.74	0.64 – 0.85
Heart failure exacerbation	0.71	0.61 – 0.83
All-cause hospitalization	0.88	0.78 – 1.00
Cardiovascular mortality	0.79	0.68 – 0.89
Stroke	1.12	0.86 – 1.54

Which of the following provides the best interpretation for the obtained results?

- ☐ A) Beta-blockers decrease both all-cause mortality and cardiovascular mortality

- ☐ B) Beta-blockers predispose to a stroke
- ☐ C) Beta-blockers affect all-cause mortality due to decreased risk of myocardial infarction
- ☐ D) Beta-blockers may exacerbate heart failure but they decrease cardiovascular mortality
- ☐ E) Beta-blockers protect from myocardial infarction but do not affect the risk of stroke

Correct Answers: 1) D 2) A 3) E

Explanation :

Relative risk and odds ratio (discussed in previous sections) are measures of association which provide point estimates of effect. They are useful in describing the *magnitude* of an effect. For example, relative risk of 2.0 indicates that the risk of an outcome in the exposed group is twice that in the unexposed group. Since relative risk and odds ratio are point estimates obtained from a random sample of the population, we need some measure of random error reported along with the point estimate. The 95% confidence interval (CI) serves this function by providing an interval of values within which we can be 95% confident that the true relative risk or odds ratio lies after accounting for random error. For example, if a relative risk of 2.0 is reported along with a 95% CI of 1.5-2.5, we can be 95% confident that the true relative risk in the population lies somewhere between 1.5 and 2.5. As previously described, a value of 1.0 for the relative risk or odds ratio indicates that there is no association between the exposure and outcome. If the 95% CI for a reported relative risk or odds ratio does not include 1.0, then there is a < 5% chance that the observed association is due to chance. Therefore, the calculated p value for such an association would be < 0.05. If the 95% CI does include 1.0, then there is a > 5% chance that the observed association is due to chance (p value is > 0.05), and the null hypothesis (no association) is accepted.

A CI can be calculated to correspond with the mean of any continuous variable. To calculate the CI around the mean you must know the following: the mean, standard deviation (SD), z-score and sample size (n). First of all, standard error of the mean (SEM) is calculated using the following formula: $SEM = SD/\sqrt{n}$. Please note that the sample size is a part of the calculation; the bigger the sample size, the tighter the CI!

The next step is to multiply the SEM with the corresponding z-score: for 95% CI it is 1.96 (remember the normal distribution and the fact that 95% of the observations lie within two standard deviations from the mean) and for 99% CI it is 2.58.

The final step is to obtain the confidence limits as shown below:

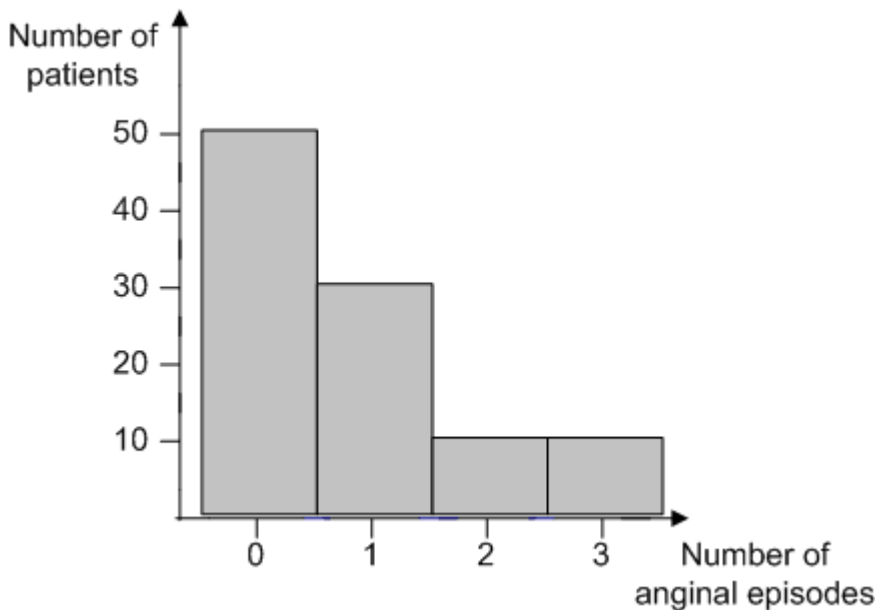
$$\text{Mean} \pm 1.96 * SD/\sqrt{n}.$$

As noted above, the width of the CI is inversely related to sample size: increasing the sample size decreases the CI, indicating higher precision of the dataset. This is demonstrated in Question #1: both studies that link OCP use with liver mass report relative risks of similar magnitude. However, study B has a wider CI which includes the value 1.0. Therefore study B has a p value > 0.05 and does not reach statistical significance. The explanation for the wider CI in study B is a smaller sample size compared to study A.

Measures of Central Tendency

Question: 1 of 3 [Qid : 21]

In an experimental study, patients suffering from stable angina are treated with a new beta-blocker. The number of anginal episodes experienced by the patients on the thirtieth day of treatment is shown in the table below.



Based on these data, what is the average number of anginal episodes experienced by patients treated with the new drug?

- ☐ A) Between 0 and 1
- ☐ B) 1
- ☐ C) Between 1 and 2
- ☐ D) 2
- ☐ E) Between 2 and 3

Question: 2 of 3 [Qid : 22]

An ICU patient has an intraarterial canula placed after cardiac surgery to monitor systolic blood pressure (SBP). Twenty four SBP values are recorded over a period of six hour, with a maximum value of 141 mmHg and a minimum value of 96 mmHg. If the next SBP recording is 200 mmHg, which of the following is most likely to remain unchanged?

- ☐ A) Mean
- ☐ B) Mode
- ☐ C) Range
- ☐ D) Variance
- ☐ E) Standard deviation

Question: 3 of 3 [Qid : 23]

A patient with severe heart failure is placed in the ICU and undergoes invasive hemodynamic monitoring. Over the next hour, the recorded values of his pulmonary artery wedge pressure are 26 mmHg, 20 mmHg, 20 mmHg, 27 mmHg, 14 mmHg and 27 mmHg. Which of the following is the median of the recorded values?

- ☐ A) 20
- ☐ B) 22
- ☐ C) 23
- ☐ D) 24
- ☐ E) 26

Correct Answers: 1) A 2) B 3) C

Explanation :

Measures of central tendency in a dataset include mean, mode and median.

Mean: To find the mean of a dataset, first, you add the values of all observations in the data set and then divide that total by the number of observations. For example, to answer Question #1, first we sum up all of the anginal episodes in study subjects:

$$0*50 + 1*30 + 2*10 + 3*10 = 80.$$

Next we divide this value by the number of patients in the study. The overall sample size is 100 (50, 30, 10, 10).

$$80/100 = 0.8.$$

We can conclude that patients experienced on average 0.8 anginal episodes on the thirtieth day of the study.

Median: The median of a dataset is the observed value that equally divides the right and left halves of the dataset. For example, if there are 13 observed values in a data set, then the median would be the value for which six of the other observed values are larger and six are lower. If the number of observations is even, then the median value is obtained by adding together the middle two values and dividing by two (see the graph below for Q3).



Fig.3. Median of a dataset is the number that divides the right half of the data from the left half.

Therefore, in this Question #3, the median is equal to $(20+26)/2 = 23$.

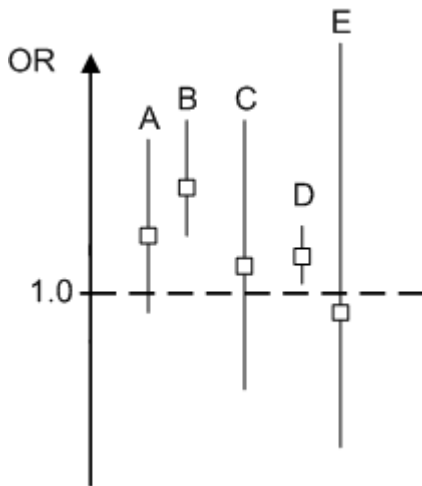
Mode: The mode is the most frequent value of the dataset.

Outlier: An outlier is defined as an extreme and unusual value observed in a dataset. It may be the result of a recording error, a measurement error, or a natural phenomenon. The mean value is typically shifted more greatly by an outlier than is the median value. The mode is not affected by an outlier.

Measures of Dispersion

Question: 1 of 2 [Qid : 24]

Four separate studies are undertaken to assess the risk of acute coronary syndrome in post-menopausal women taking hormone replacement therapy. The results of the individual studies as well as the result of a meta-analysis are shown on the table below. Each study result is presented as an odds ratio along with a confidence interval. Which of the following results most likely corresponds to the meta-analysis?



- ☐ A) A
- ☐ B) B
- ☐ C) C
- ☐ D) D
- ☐ E) E

Question: 2 of 2 [Qid : 25]

A study addresses the role of air pollution in asthma development. 100 children with diagnosed asthma and 200 children without asthma are asked a series of questions regarding their homes. An air pollution index ranging from 0 to 10 is then calculated based on each child's responses. The mean air pollution index for children with asthma is calculated as 4.3 (95% confidence interval 3.1 – 5.5). Which of the following statistical changes would be most likely if more asthmatic children were included in the study?

- | | Standard error of the mean | Upper confidence limit | Lower confidence limit |
|------------------------------------|----------------------------|------------------------|------------------------|
| ☐ A) ↑ | ↓ | ↓ | ↓ |
| ☐ B) ↓ | ↓ | ↓ | ↑ |
| ☐ C) ↓ | ↓ | ↓ | ↓ |
| ☐ D) ↓ | ↓ | ↑ | ↓ |
| ☐ E) No change | ↓ | ↓ | ↑ |

Correct Answers: 1) D 2) B

Explanation :

Range, standard deviation, standard error of the mean, and percentile are all measures of dispersion (or variability).

Range: Represents the difference between the highest and lowest value in the dataset.

Standard deviation (SD) measures dispersion around the mean in the study sample whereas standard error of the mean (SEM) shows how precisely the sample represents the study population. SEM is always smaller than SD because it is calculated as SD divided by the square root of sample size!

SD is calculated as follows:

$$SD = \sqrt{\frac{\sum (x - X)^2}{n - 1}}$$

Where

SD represents standard deviation

sum; means the sum of all values

X represents the mean

x represents the individual values in the data set

n represents the number of data points in the set

Note that n is inversely related to SD. In other words, as the number of data points in the set increases, the standard error of the mean decreases. As noted in the section on confidence intervals, the formula for confidence intervals is as follows:

$$95\% \text{ CI} = \text{Mean} \pm 1.96 \times \text{SD} / \sqrt{n}.$$

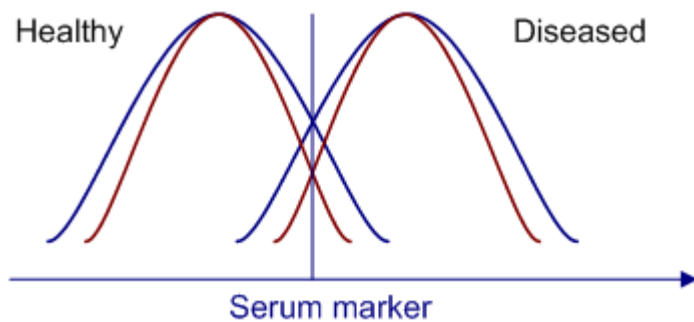
In other words, confidence intervals vary directly with SD and inversely with the sample size. In other words, as the sample size increases, the confidence interval decreases (narrows). Apply this principle to Question #1. A meta-analysis contains more data points than any of the individual studies from which it is derived. Since the sample size is larger in the meta-analysis, the confidence interval will be narrower. Hence, the correct choice is D. Also apply this principle to Question #2. As the number of data points in the set increases (number of asthmatic children), the SEM decreases and the confidence interval narrows (Choice B).

Percentile describes the percentage of population below a specific value. For example, if your score on the exam corresponds to 80th percentile, then only 20% of examinees scored above you. Interquartile range is the difference between the values corresponding to the 75th and 25th percentile..

Sensitivity and Specificity

Question: 1 of 6 [Qid : 26]

A new test has been developed for early diagnosis of pancreatic cancer. It uses a serum marker level as an indicator of the neoplastic process. The graph below demonstrates the distribution of serum marker levels in both healthy and diseased populations.



Compared to the blue curves, the red curves are associated with:

- ☐ A) Higher sensitivity and lower specificity
- ☐ B) Higher sensitivity and higher specificity
- ☐ C) Higher sensitivity and same specificity
- ☐ D) Lower sensitivity and higher specificity
- ☐ E) Lower sensitivity and lower specificity

Question: 2 of 6 [Qid : 27]

A new diagnostic test for tuberculosis has a sensitivity of 90% and a specificity of 95%. If applied to a population of 100,000 patients in which the prevalence of tuberculosis is 1%, how many false negative results would you expect?

- ☐ A) 10
- ☐ B) 50
- ☐ C) 100
- ☐ D) 500
- ☐ E) 900
- ☐ F) 1,000
- ☐ G) 9,000

Question: 3 of 6 [Qid : 28]

A rare disorder of amino acid metabolism causes severe mental retardation if left untreated. If the disease is detected soon after birth a restrictive diet prevents mental abnormalities. Which of the following characteristics would be most desirable in a screening test for this disease?

- ☐ A) High Sensitivity
- ☐ B) High Specificity

- ☐ C) High Positive predictive value
- ☐ D) High Cutoff value
- ☐ E) High Accuracy

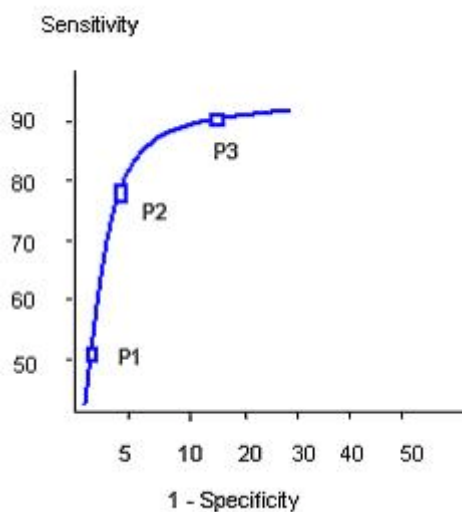
Question: 4 of 6 [Qid : 29]

A rapid test that is used to diagnose HSV infection is positive in HSV-infected patients 9 times more often than in non-infected patients. Which of the following expressions is used to derive this information?

- ☐ A) True positives/All positives
- ☐ B) True positives/True negatives
- ☐ C) Sensitivity/Specificity
- ☐ D) Sensitivity/(1 – Specificity)
- ☐ E) Specificity/(1 – Sensitivity)

Question: 5 of 6 [Qid : 30]

A new serum marker shows promise in the early diagnosis of colon cancer. It represents a fetal antigen that has minimal expression in healthy adults, but has increased expression in those with colon cancer. Various serum concentration levels (P1, P2, and P3) are tested as cutoff points for diagnosis of disease. The sensitivity and specificity of the test at each of these serum concentrations is then compared to the gold standard (excisional biopsy). The following curve is constructed.



Which of the following is the best statement concerning this new test?

- ☐ A) P1 represents the cutoff point with the best 'ruling out' possibility
- ☐ B) P2 represents the cutoff point with the best 'ruling in' possibility
- ☐ C) P3 corresponds to the cutoff point with the highest positive predictive value
- ☐ D) P3 corresponds to a lower serum marker value than does P1
- ☐ E) The higher the serum marker level used as a cutoff point, the lower the specificity

Question: 6 of 6 [Qid : 31]

A 38-year-old Caucasian primigravida presents to your office at 20 weeks' gestation for prenatal counseling. She is concerned about the risk of Down syndrome and asks about methods of early diagnosis. You explain that triple screening may detect up to 50% of cases and amniocentesis may detect up to 90%. She decides not to undergo either test and gives birth to a child with Down syndrome. While comparing both tests during patient counseling you specifically emphasized:

- ☐ A) Increased false negatives
- ☐ B) Increased false positives
- ☐ C) Increased positive predictive value
- ☐ D) Increased negative predictive value
- ☐ E) Increased sensitivity

Correct Answers: 1) B 2) C 3) A 4) D 5) D 6) E

Explanation :

Sensitivity and specificity are measures of a diagnostic test's validity. Sensitivity is defined as the proportion of diseased subjects who test positive for disease. Specificity is defined as the proportion of disease-free subjects who test negative for disease.

Consider the following 2 x 2 table:

Test results	Disease Present	Disease Absent	Total
Positive	A True positive (TP)	B False positive (FP)	A+B
Negative	C False Negative (FN)	D True Negative (TN)	C+D
Total	A+C	B+D	A+B+C+D

Sensitivity = $TP/(TP+FN)$ or $A/(A+C)$.

Sensitivity represents the probability of testing positive in patients having the disease. For example, sensitivity of 90% means that 90 of 100 patients with the disease would test positive. Question #2 presents a population of 100,000 with a reported tuberculosis incidence of 1%. In this population there are therefore 1,000 cases of existing tuberculosis. The new diagnostic test which has a sensitivity of 90% would identify 900 cases but would not identify the disease in the remaining 100 cases (false negatives). A test with a high sensitivity is typically used as a screening test because it can 'rule in' as many people with the disease as possible. In Question #3 it is essential to diagnose as many patients with the hereditary metabolic disease as possible because (1) the condition has severe complications and (2) it is potentially treatable if diagnosed early. Therefore, a screening test with a high sensitivity is important.

Specificity = $TN/(TN+FP)$ or $D/(B+D)$

Specificity represents the probability of testing negative in patients without the disease. Question #2 presents a population of 100,000 with a reported tuberculosis incidence of 1%. In this population, there are therefore 99,000 people free of the disease. The new test would be negative in 95% of these people (94,050)

but would be false positive in the remaining 4,950 people. A test with a high specificity is typically used as a confirmatory test because it can 'rule out' as many people without the disease as possible.

A diagnostic test with perfect validity would have sensitivity and specificity equal to 1, but this is seldom possible. Typically, there is a trade-off between sensitivity and specificity. Imagine a serum marker used in the diagnosis of an oncologic disease (as in Question #1). If the serum level of the marker is measured in healthy and diseased individuals, there is almost always an overlap between healthy individuals with 'high-normal' values and diseased individuals with 'low-abnormal' values (see Fig.4). If the cutoff point is set at point X, the right tail of the 'healthy' curve represents false positives and the left tail of the 'diseased' curve represents false negatives.

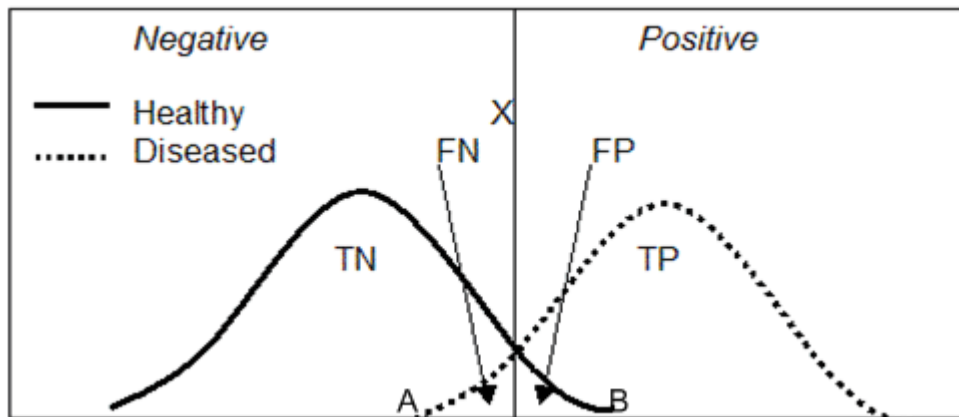


Fig. 4. The bell curves in the above diagram represent the distribution of serum marker levels in the healthy and diseased population. X represents the cutoff value for positive and negative test results. Point A corresponds to 100% sensitivity and point B corresponds to 100% specificity.

Shifting the cutoff value towards point A increases sensitivity but decreases specificity. Shifting the cutoff value towards point B decreases sensitivity but increases specificity. Decreased overlap between the healthy and diseased population curves as demonstrated by the red curves (compared to the blue curves) in Question #1, decreases both the number of false positives and false negatives. Therefore the red curves are associated with higher sensitivity and specificity.

The curve shown in Question #5 is called a receiver operating characteristic (ROC) curve. It illustrates the tradeoff between sensitivity and specificity which is made when choosing a cutoff value for positive and negative test results. In this example, the P3 cutoff point shows high sensitivity and low specificity, while the P1 cutoff point shows a low sensitivity and high specificity. Based on these observations, it can be concluded that P3 corresponds to a lower serum marker value than does P1.

The area under ROC represents accuracy of the test (the number of true positives plus true negatives divided by the number of all observations). An accurate test would have area under the ROC close to 1.0 (rectangular shape) whereas a test with no predictive value would be represented by a straight line (see Fig. 5).

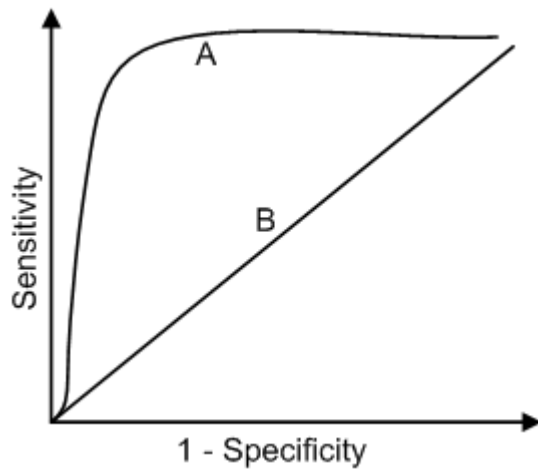


Fig. 5. Two receiver operating characteristic (ROC) curves are shown. Curve A has area under the curve close to 1.0 and represents an accurate test. Curve B has area under the curve of 0.5 and lacks predictive value.

Another important indicator of test performance is the likelihood ratio. The positive likelihood ratio is calculated by dividing sensitivity by (1-specificity). A positive likelihood ratio of 9 indicates that a positive test result is seen 9 times more frequently in patients with the disease than in patients without the disease. Unlike predictive values, the likelihood ratio is independent of disease prevalence.

Predictive Values

Question: 1 of 6 [Qid : 32]

A new stool test for *H. pylori* infection yields positive results in 80% of infected patients and in 10% of uninfected patients. Prevalence of *H. pylori* infection in the population is 10%. What is the probability that a patient who tests positive with the new test is infected with *H. pylori*?

- ☐ A) 25%
- ☐ B) 33%
- ☐ C) 47%
- ☐ D) 54%
- ☐ E) 75%

Question: 2 of 6 [Qid : 33]

A 52-year-old Caucasian female presents to your office with a self-palpated thyroid nodule. After the appropriate work-up, fine-needle aspiration (FNA) of the nodule is performed. The FNA result is negative. As you are explaining the test result, the patient asks, "What are the chances that I really do not have cancer?" You reply that the probability of thyroid cancer is low in her case because FNA has a high:

- ☐ A) Specificity
- ☐ B) Sensitivity
- ☐ C) Positive predictive value
- ☐ D) Negative predictive value
- ☐ E) Validity

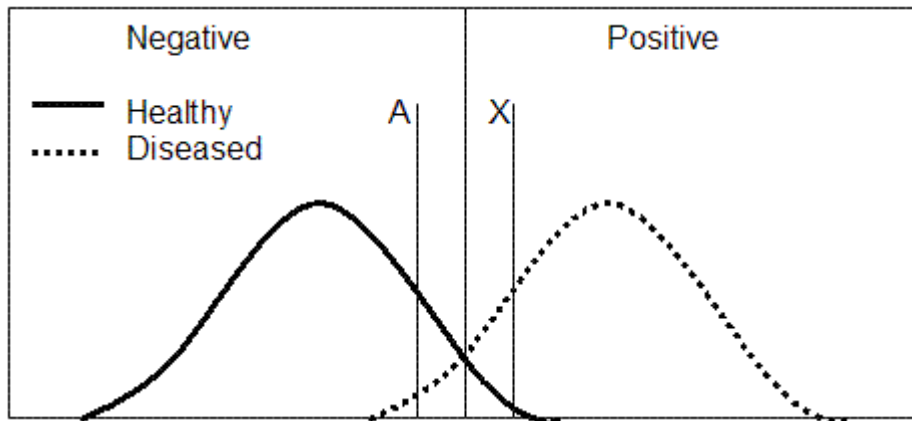
Question: 3 of 6 [Qid : 34]

A serologic test is introduced for the diagnosis of hepatitis C virus (HCV) infection. When tested on the general population, the sensitivity and specificity of the test are 85% and 78%, respectively. If the test is applied to a population of IV drug abusers with a higher probability of HCV infection, which of the following changes would you expect?

- | | Specificity | Positive Predictive Value | Negative Predictive Value |
|--|-------------|---------------------------|---------------------------|
|--|-------------|---------------------------|---------------------------|

Question: 4 of 6 [Qid : 35]

A new test for early detection of ovarian cancer is under investigation. It measures a serum marker level as an indicator of the neoplastic process. The results of the study demonstrate that the serum marker level is correlated with the presence of ovarian cancer in the women under study.



If the cutoff point is moved from X to A, the positive predictive value will:

- ☐ A) Decrease
- ☐ B) Increase
- ☐ C) Remain unchanged
- ☐ D) Cannot be determined based on the data provided

Question: 5 of 6 [Qid : 36]

190 patients with exercise-induced chest pain and a normal baseline ECG undergo stress ECG followed by coronary angiography. Coronary angiography is interpreted as positive if at least one of coronary arteries has an atherosclerotic lesion with $\geq 70\%$ luminal stenosis. The following results are obtained (see the table below).

ECG Stress Test	Coronary angiography	
	Positive	Negative
Positive	90	10
Negative	12	78

According to the study results, if a patient with exercise-induced chest pain has a negative ECG stress test, what is his/her probability of having a positive result on coronary angiography?

- ☐ A) 10%
- ☐ B) 11%
- ☐ C) 12%
- ☐ D) 13%
- ☐ E) 15%

Question: 6 of 6 [Qid : 37]

Several tests have been developed to measure serologic markers of breast cancer. The sensitivity and specificity for diagnosis of early stage breast cancer vary from test to test. If positive, which of the following tests will have the highest predictive value for the disease?

- ☐ A) Sensitivity - 80%, specificity - 90%
- ☐ B) Sensitivity - 65%, specificity - 97%
- ☐ C) Sensitivity - 70%, specificity - 94%
- ☐ D) Sensitivity - 75%, specificity - 92%
- ☐ E) Sensitivity - 85%, specificity - 90%

Correct Answers: 1) C 2) D 3) B 4) A 5) D 6) B

Explanation :

Predictive values are important measures of the post-test probability of disease.

Consider the following two-by-two table:

Test results	Disease Present	Disease Absent	Total
Positive	A True positive (TP)	B False positive (FP)	A+B
Negative	C False Negative (FN)	D True Negative (TN)	C+D
Total	A+C	B+D	A+B+C+D

Positive predictive value (PPV) represents the probability of having the disease if the test is positive. It is calculated using the following formula:

$$PPV = TP/(TP + FP) = A/(A+B)$$

Negative predictive value (NPV) represents the probability of being free of the disease if the test is negative. It is calculated using the following formula:

$$NPV = TN/(TN+FN) = D/(C+D)$$

Unlike sensitivity, specificity and likelihood ratios, predictive values depend on the prevalence of the disease in the population tested. If the prevalence is high, a positive test is more likely to be a true positive (PPV is high). If the prevalence is low, a negative test is more likely to be a true negative (NPV is high).

It is also important to understand that predictive values are impacted by the pre-test probability of disease. In patients with a high pre-test probability of disease, the PPV of diagnostic testing is increased. Imagine performing HIV testing on two patients. The first patient has multiple risk factors for infection and therefore has a high pre-test probability of HIV. The second patient has no risk factor for infection and therefore has a low pre-test probability of the disease. A positive result in the first patient has

a higher PPV (post-test probability of the disease) than a positive result in the second patient, although sensitivity and specificity of the HIV test are the same for both patients.

It is possible to calculate predictive values if given the sensitivity, specificity and disease prevalence. Bayes theorem, an important theorem in probability theory is used for calculations.

Applying Bayes theorem to Question #1:

Sensitivity is 80% (0.8) and specificity is 90% (0.9). Prevalence of the disease is 10% (0.1). To calculate the predictive values, begin by calculating the probability of obtaining a true positive: multiply sensitivity by prevalence (0.8×0.1). Then, calculate the probability of obtaining a false positive: multiply (1-specificity) by (1-prevalence) (0.1×0.9). According to the definition, PPV equals the number of true positives divided by the total number of positive test results. Therefore, PPV is equal to $(0.8 \times 0.1) / [(0.8 \times 0.1) + (0.1 \times 0.9)] = 47\%$. A similar method can be used to calculate NPV.

Another way of solving Question #1 is by plugging in numbers. Imagine that the population consists of 100 patients. Since the disease prevalence is 10%, that means 10 patients have the disease and 90 do not. Performing a test with 80% sensitivity on 10 diseased patients yields 8 true positive. Performing a test with 90% specificity on 90 patients without disease yields 9 false positives. PPV equals the fraction of true positives divided by all positives. Therefore, PPV in this case is equal to $8 / (8 + 9) = 47\%$.

Question #5 asks for the reciprocal of NPV: what is the probability of having the disease (positive coronary angiogram) if you have a negative test (EKG stress test)? It can be calculated as the following:

$$(1 - \text{NPV}) = 1 - D / (C + D) = C / (C + D) = 12 / (12 + 78) = 0.13 \text{ (13\%)}$$

The cutoff value of a test determines the balance between false positives and false negatives. It therefore affects the sensitivity and specificity of a test (see the discussion in section 9). In turn, specificity of a test is an important determinant of PPV, because a high specificity is associated with fewer false positives (Question #6). In Question #4, moving the cutoff value from point X to point A increases sensitivity and therefore also increases the number of true positives. At the same time, this move also decreases the specificity and therefore increases the number of false positives. Because the disease prevalence is low (i.e. there are more healthy than diseased individuals in the population), the increase in false positives from moving the cutoff point in this manner is larger than the increase in true positives. The overall result is a decrease in the positive predictive value.

Screening

Question: 1 of 2 [Qid : 38]

A new screening test is being evaluated for the early detection of stomach cancer. The test relies on measurement of a new serologic marker for gastric adenocarcinoma. The study concludes that, compared to the traditional strategy of endoscopic evaluation of high-risk patients, the new screening test increases survival by several weeks. This increase in survival is statistically significant, although no difference is detected in the rate of radical gastrectomy between two groups. Which of the following is most likely to affect the study results presented above?

- ☐ A) Low sensitivity
- ☐ B) Selection bias
- ☐ C) Lead-time bias
- ☐ D) Confounding
- ☐ E) Recall bias

Question: 2 of 2 [Qid : 39]

A new screening test for prostate cancer tends to diagnose non-aggressive forms of the disease but often misses more aggressive forms. An apparent increase in survival after implementation of the test would be most likely affected by:

- ☐ A) Confounding
- ☐ B) Length-time bias
- ☐ C) Selection bias
- ☐ D) Ascertainment bias
- ☐ E) Measurement bias

Correct Answers: 1) C 2) B

Explanation :

Lead-time bias: The goal of a screening test is to detect the disease early enough to allow for successful intervention and to improve the outcome. Therefore, two components of a useful screening test should be emphasized: 1) early detection of a disease (earlier than routine diagnostics) and 2) increase in survival associated with the implementation of the test. Sometimes a screening test leads to earlier detection of a disease and to an apparent increase in survival, yet when the data is scrutinized more closely it is found that the apparent increase in survival is due only to earlier detection and not to successful intervention or improved prognosis. This phenomenon is referred to as lead-time bias (see Fig. 6). For example, in Question #1 the new test appears to detect the disease earlier than the traditional approach but survival only increases by several weeks and the rates of radical gastrectomy are unchanged. The explanation for the apparent increase in survival is early diagnosis, not successful treatment of stomach cancer; prognosis seems to be the same for both groups.

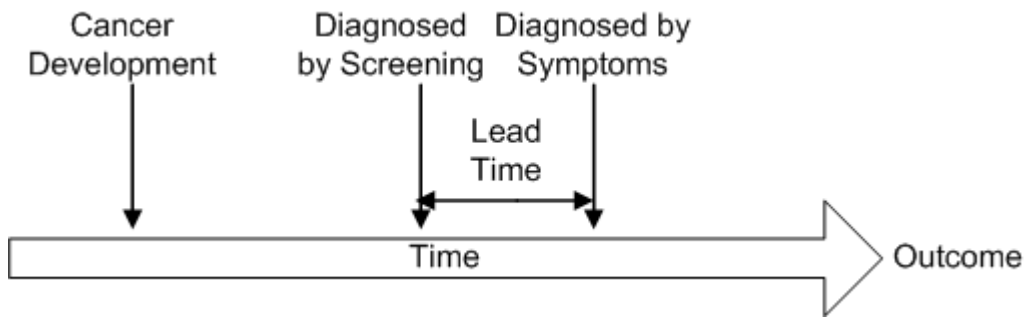


Fig.6. Lead time represents the time difference between the detection of cancer by a screening test and the time of diagnosis by disease symptoms or by a prior method of diagnosis.

Length-time bias: Length-time bias is a phenomenon whereby a screening test preferentially detects less aggressive forms of a disease and therefore increases the apparent survival time. This is the case in Question #2, where a new screening test detects more non-aggressive prostate cancers and fewer aggressive ones than the previous method of diagnosis.

Study Design

Question: 1 of 5 [Qid : 40]

An investigator suspects that acetaminophen use during the first trimester of pregnancy can cause neural tube defects. He estimates the general population risk of having neural tube defect is 1:1,000. Which of following is the best study design to investigate the hypothesis?

- ☐ A) Cohort Study
- ☐ B) Case Control Study
- ☐ C) Clinical Trial
- ☐ D) Ecologic Study
- ☐ E) Cross-Sectional Study

Question: 2 of 5 [Qid : 41]

A group of investigators are studying the relationship between a particular 5-lipoxygenase genotype and atherosclerosis. A study population is randomly selected. Blood samples are obtained for leukocyte genotyping, and ultrasonography is performed to assess carotid intima-media thickness, a marker of atherosclerosis. It is then concluded that the particular 5-lipoxygenase genotype is associated with a predisposition to atherosclerosis. Which of the following choices identifies the study design used by the investigators?

- ☐ A) Case Series Report
- ☐ B) Cohort Study
- ☐ C) Case-Control Study
- ☐ D) Cross-Sectional Study
- ☐ E) Randomized Clinical Trial

Question: 3 of 5 [Qid : 42]

Officials at a large community hospital report an increased incidence of acute lymphocytic leukemia (ALL) among children aged 5-12. They point out that some households in the community are exposed to chemical waste from a nearby factory. They believe that chemical waste causes leukemia. If a study is designed to evaluate the hospital officials' claim, which of the following subjects are most likely to comprise the control group?

- ☐ A) Children exposed to the chemical waste who do not suffer from ALL
- ☐ B) Children not exposed to the chemical waste who do not suffer from ALL
- ☐ C) Children from the outpatient clinic who do not suffer from ALL
- ☐ D) Children not exposed to the chemical waste who suffer from ALL
- ☐ E) Children who suffered from ALL but got cured

Question: 4 of 5 [Qid : 43]

500 women aged 40-54 who present for routine check-ups are asked about their meat consumption. 20% of the women turn out to be vegetarian. During the ensuing 5 years, 5 vegetarians and 43 non-vegetarians

develop colorectal cancer. Which of the following best describes the study design?

- ☐ A) Case Series Report
- ☐ B) Cohort Study
- ☐ C) Case-Control Study
- ☐ D) Cross-Sectional Study
- ☐ E) Randomized Clinical Trial

Question: 5 of 5 [Qid : 44]

A group of researchers wants to investigate an outbreak of acute diarrhea that occurred in a small coastal town. About 50 people developed severe hemorrhagic diarrhea and one fatal case was reported. The researchers believe that the outbreak is related to the seafood prepared at one of the coastal restaurant. Which of the following study designs is most appropriate to investigate the hypothesis?

- ☐ A) Cohort study
- ☐ B) Cross-sectional study
- ☐ C) Case-control study
- ☐ D) Ecologic study
- ☐ E) Clinical trial

Correct Answers: 1) B 2) D 3) C 4) B 5) C

Explanation :

A useful algorithm for determining study design is shown in Fig.7.

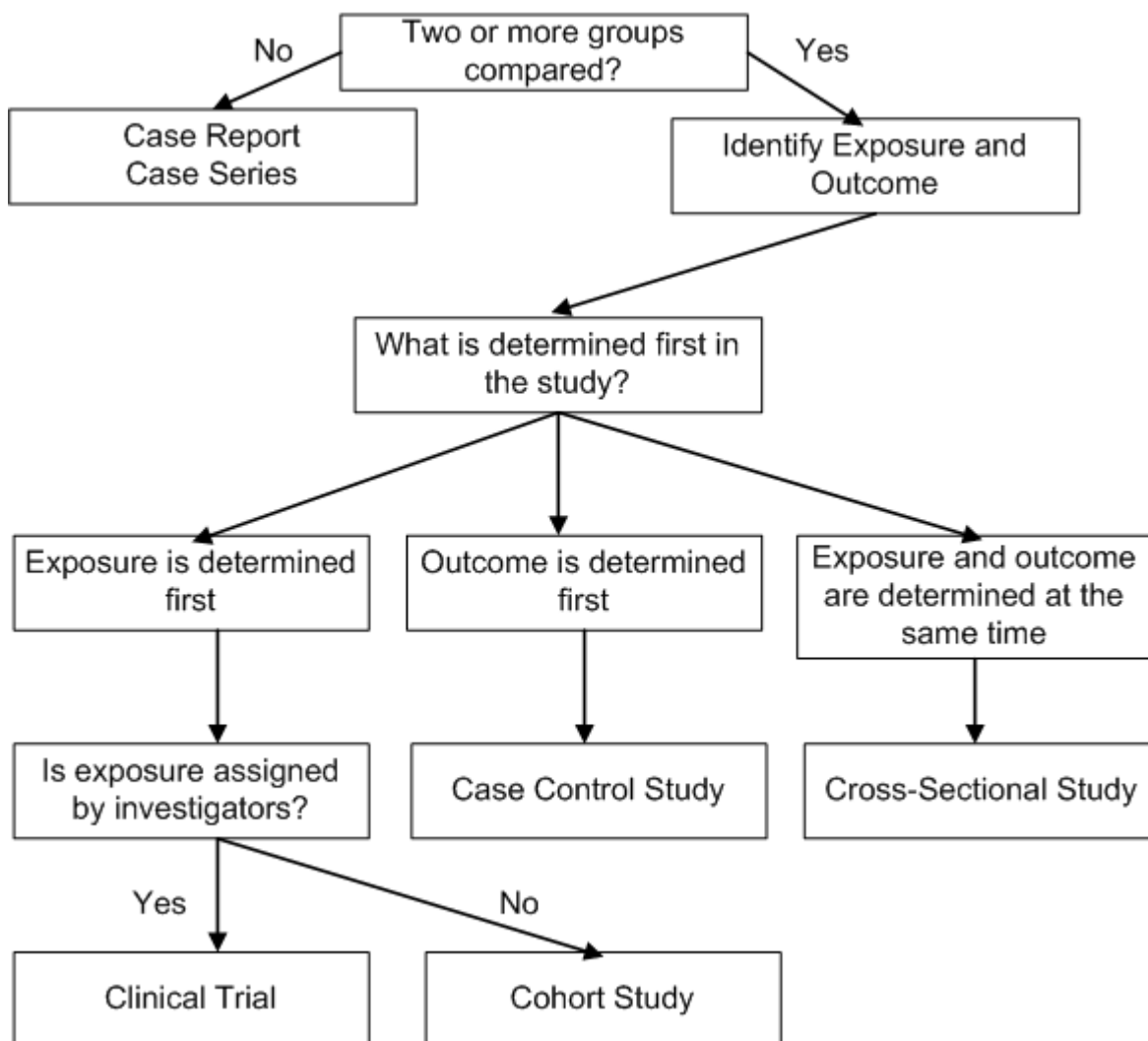


Fig.7. An algorithm to determine study design.

Once investigators formulate the hypothesis they would like to test, they should define the study population and determine the study design that best fits the hypothesis.

From the perspective of general epidemiology, studies can be classified as descriptive and analytical (see table 1). Descriptive studies are used to outline disease distribution in the population; they do not directly address causality. Analytical studies are used to determine the cause of the disease.

Descriptive studies	Analytical Studies
- Individual-level - Case Reports - Case Series - Cross-sectional studies - Population-level - Correlational (ecologic)	- Observational Studies - Case-Control Studies - Cohort Studies - Interventional Studies - Randomized Clinical trials

Table1. Common study designs.

Descriptive studies: Descriptive studies include case reports, case series, cross-sectional studies, and correlational (ecologic) studies. Case reports and case series provide description of individual patient cases or a group of cases sharing the same diagnosis. Typically, case reports and case series describe unusual cases that may provide greater understanding of the disease or that may have public health significance. For example, case reports about young men suffering from pneumocystis pneumonia led to the discovery of a new disease entity called AIDS. A cross-sectional study (prevalence study) is characterized by the simultaneous measurement of exposure and outcome. It is a snapshot study design frequently used for surveys. It has the advantage of being cheap and easy to perform. Its major limitation is the fact that a temporal relationship between exposure and outcome is not always clear, although in Question #2 demonstrating a temporal relationship was easy since acquiring a particular genotype definitely precedes atherosclerosis. A correlational study (ecologic or aggregate study) deals with information on a population level rather than on an individual level. Example: a steady decline in cigarette sales over the past several decades is associated with a decline in the incidence of ischemic heart disease during the same period. The major limitation with correlational studies is the potential for erroneous conclusions regarding the exposure-disease relationship on an individual level drawn from the population-level information. This type of erroneous conclusion is called 'ecologic fallacy'.

Analytical studies: Analytic studies include observational studies (case-control, cohort) and interventional studies such as randomized clinical trials.

Case-control studies address the exposure-disease relationship by comparing the exposure status in cases (diseased patients) with controls (non-diseased patients). Therefore, the direction of the investigation is retrospective: find subjects with the disease and find appropriate control subjects without the disease. Then determine the previous exposure status of both groups and compare the exposure status in cases and controls. Case-control studies are easier to organize and conduct than cohort studies and they are much cheaper. Case-control studies are the preferred study design for small infectious outbreaks and for rare diseases. For example, case-control studies suggested a possible association between Reye syndrome and aspirin use in children. In Question #1, investigators want to investigate the potential cause (acetaminophen) of a rare outcome (neural tube defects) and therefore a case-control study is appropriate. In Question #5, health authorities want to investigate an outbreak of infectious diarrhea. They identify 50 patients (cases) affected by the disease. The next step would be to select people from the town population who are not affected by the disease (controls). Once cases and controls are selected, investigators should inquire about their recent restaurant visits (exposure) and, finally, the exposure status should be compared in cases and controls. Unlike cohort studies, patients are not followed over time for the development of the disease and therefore case-control studies do not directly determine the risk of the disease based on exposure. The measure of association in case-control studies is exposure odds ratio (see section 2 for measures of association) that compares the odds of exposure in cases with the odds of exposure in controls. It is important to understand the role of the control group in case-control studies. Selection of control subjects is intended to provide the estimation of exposure frequency among the population; this exposure frequency then is compared to that of cases. Therefore, a proper selection of control subjects underlies the quality of the study. In Question #3, children from the outpatient clinic that serves the community may be good candidates for the control group. Selecting controls based on exposure status is inappropriate because comparing the exposure status in cases and controls underlies the analysis.

Cohort studies are designed by selecting a group of subjects free of the disease of interest. This group (cohort) typically shares a common experience (e.g., women of a certain age who come for routine check-up). Exposure status (a potential risk-factor) is determined in these individuals at the beginning of the study, and the cohort is then followed over time for development of the disease of interest. In Question #4 a typical cohort study is described. 500 disease-free women are selected and their exposure status (vegetarian vs. non-vegetarian) is determined. Then they are followed over 5 years for the development of colorectal cancer.

The most famous cohort study ever conducted is the Framingham heart study. This study identified the major risk factors for cardiovascular disease such as hypercholesterolemia, diabetes, smoking and hypertension. Unlike case-control studies, cohort studies are designed to describe the risk of the disease directly (the probability of developing the disease over a certain period of time based on risk factors). A relative risk is calculated based on the data which compares the risk of the disease in exposed subjects to the risk of the disease in unexposed subjects (see section 2 for measures of association). The cohort can be followed for the development of an outcome prospectively (so called prospective or concurrent cohort studies) or retrospectively (so called retrospective or non-concurrent cohort studies).

The term 'longitudinal study' applies to studies that follow study subjects over a long period of time, typically many years. The Framingham heart study is an example of a longitudinal cohort study.

Clinical trials are similar to cohort studies in terms of a prospective study design. Unlike cohort studies, they do not simply record the exposure at the baseline. Rather, exposure is assigned to study subjects. Therefore clinical trials are called interventional (experimental) as opposed to observational. Exposure may be in the form of a drug, vaccine, or intervention. Once the exposure status is assigned, patients are followed over time to determine the outcome or end-point. End-points are specified in advance and can be subdivided into primary (of primary importance) and secondary. Examples of end-points in clinical trials are all-cause mortality, myocardial infarction, hospitalization, etc. The results are typically reported in terms of relative risk.

A very common type of analysis employed in prospective studies is survival analysis (time-to-event analysis) discussed separately.

Selection and Measurement Bias

Question: 1 of 5 [Qid : 45]

A study is conducted to assess the relationship between ethnicity and end-stage renal disease. Two groups of pathologists independently study specimens from 1,000 kidney biopsies. The first group of pathologists is aware of the race of the patient from whom the biopsy came, while the second group is blinded as to the patient's race. The first group reports 'hypertensive nephropathy' much more frequently for black patients than the second group. Which of the following types of bias is most likely present in this study?

- ☐ A) Confounding
- ☐ B) Nonresponse bias
- ☐ C) Recall bias
- ☐ D) Referral bias
- ☐ E) Observer bias

Question: 2 of 5 [Qid : 46]

A cohort study is conducted to assess the relationship between a high-fat diet and colorectal adenocarcinoma. The study shows that no association exists between the exposure and the outcome after controlling for known risk factors (age, fiber consumption, and family history of cancer): relative risk - 1.35 ($p = 0.25$). The investigators also report that 40% of the subjects in the high-fat group and 36% of those in the low-fat group were lost to follow-up. Based on this information, which of the following biases is most likely to be present?

- ☐ A) Observer bias
- ☐ B) Selection bias
- ☐ C) Ascertainment bias
- ☐ D) Recall bias
- ☐ E) Confounding

Question: 3 of 5 [Qid : 47]

A study is conducted to assess the relationship between the use of an over-the-counter pain reliever during pregnancy and the development of neural tube defects in offspring. Mothers whose children have neural tube defects and age-matched controls with unaffected children are interviewed using a standard questionnaire. The study shows that use of the pain reliever during pregnancy increases the risk of neural tube defects, even after adjusting for race, other medications, family history of congenital abnormalities and serum folate level: OR = 1.5, $p = 0.03$. Which of the following biases is of major concern when interpreting the study results?

- ☐ A) Nonresponse bias
- ☐ B) Susceptibility bias
- ☐ C) Recall bias
- ☐ D) Observer bias
- ☐ E) Confounding

Question: 4 of 5 [Qid : 48]

A large-scale clinical trial is being planned to evaluate the effect of a non-selective beta-blocker, propranolol, on the clinical course of portal hypertension. The primary outcomes of the study are all-cause mortality and major gastrointestinal hemorrhage. Secondary outcomes are minor gastrointestinal hemorrhage and the number of hospitalizations. The investigators are concerned about the possibility that episodes of major gastrointestinal hemorrhage could be over-reported in the placebo group. Which of the following is the most useful technique to reduce this possibility?

- ☐ A) Randomization
- ☐ B) Blinding
- ☐ C) Matching
- ☐ D) Restriction
- ☐ E) Stratified analysis

Question: 5 of 5 [Qid : 49]

In a population with a high incidence of cardiovascular disease, diabetics are at least twice as likely to die from myocardial infarction as are non-diabetics. A case-control study conducted in the community identifies 1,000 people with sustained myocardial infarction and 1,000 people without sustained myocardial infarction. The subjects are asked whether they have a history of diabetes mellitus. According to the study results, diabetes has a protective effect against myocardial infarction. Which of the following best explains the observed study results?

- ☐ A) Latent period
- ☐ B) Selection bias
- ☐ C) Observer bias
- ☐ D) Hawthorne effect
- ☐ E) Recall bias

Correct Answers: 1) E 2) B 3) C 4) B 5) B

Explanation :

Sometimes study results describing the association between exposure and outcome can be distorted by systematic errors in the study design or analysis. These systematic errors are referred to as biases, and are distinct from the random error which comes from sampling a population. There are many potential flaws in design that can compromise the study results. The three basic types of bias are: selection bias, measurement (information) bias, and confounding (see table 2).

Selection bias: results when subjects selected for the study are not representative of the study population <i>Examples:</i> Nonresponse bias Referral Bias Susceptibility Bias Berkson Fallacy Prevalence Bias	Measurement (information) bias: results from inaccurate estimation of exposure and/or outcome <i>Examples:</i> Recall Bias Observer Bias	Confounding: results when the effect of the main exposure is mixed with the effect of extraneous factors.
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Tables 2. Types of Bias.

Selection bias results from selection of study subjects that are not representative of the study population. For example, selecting control subjects for a case-control study from hospitalized patients can potentially bias the results because the exposure frequency in hospitalized patients does not necessarily that of the general population. This type of selection bias is called Berkson fallacy. Referral bias results when patients are sampled from specialized medical centers and therefore they do not represent the general population. For example, patients in a university hospital may have more severe illness and higher mortality rates than individuals with the same condition in a community hospital. Another example of selection bias is selective loss to follow-up. This occurs in cohort studies. If people from one group (exposed or unexposed) who are lost to follow-up are more likely to develop the outcome in question than those lost to follow-up from the other group, then selection bias results. A high rate of follow-up loss creates a high potential for selection bias in prospective studies (see Question #2). Non-response bias may occur when study design allows subjects to decide whether or not to participate in the study. Imagine a health survey conducted by a random selection of phone numbers. The phone numbers selected are called and people are interviewed using a standardized questionnaire. There are always people who would refuse to participate in the survey. If the refusal is somehow related to their health status (e.g., they are sicker than the general population), then non-response selection bias results. Prevalence bias (Neyman bias) may occur when incidence of a disease is estimated based on prevalence, and data become skewed by selective survival. Question #5 describes a case of prevalence bias. Diabetics are more likely to die from myocardial infarction than are non-diabetics. If living patients who have sustained myocardial infarction are asked about their diabetes status, it is likely that diabetics will be under-represented because non-diabetics 'selectively survived' their cardiovascular events. Susceptibility bias occurs when the treatment regimen selected for a patient depends on the severity of the patient's condition. Imagine patients with acute coronary syndrome. Healthier patients may be preferentially selected for coronary intervention, while sicker patients may instead be selected for medical therapy. This may create bias whereby outcomes from coronary intervention appear superior to medical therapy simply because the subjects who underwent coronary intervention were healthier.

Measurement (information) bias results from inaccurate estimation of exposure and/or outcome. Measurement bias implies that exposure and/or outcome data are systematically misclassified (e.g., exposed cases are labeled as unexposed). Misclassification can be differential (e.g., outcome in the exposed subjects is misclassified) or non-differential (e.g., outcome in all groups is misclassified). Recall bias is a typical example of measurement bias which should always be considered as a potential problem in case-control studies. Recall bias can result in overestimation of the effect of exposure. In Question #3, the women whose children have neural tube defect are more likely to report use of the drug than women whose children are healthy. This over-reporting is due to psychological trauma induced by the birth of the baby with a congenital abnormality and search for the potential explanation of the problem.

Observer bias (ascertainment bias, detection bias or assessment bias) is a form of measurement bias that occurs when the investigator's decision is adversely affected by knowledge of the exposure status. In Question #1, some pathologists' decisions were influenced by the fact that hypertensive nephropathy is a common cause of end-stage renal disease in black patients. In Question #4, health care providers knowing

the treatment status of patients may over or under-report gastrointestinal bleeding episodes. Blinding of the health care provider is an effective tool to avoid observer bias.

Confounding Bias

Question: 1 of 4 [Qid : 50]

A case-control study is conducted to assess the association between alcohol consumption and lung cancer. 100 patients with lung cancer and 100 controls are asked about their past alcohol consumption. According to the study results, alcohol consumption is strongly associated with lung cancer (OR = 2.25). The researchers then divide the study subjects into two groups: smokers and non-smokers. Subsequent statistical analysis does not reveal any association between alcohol consumption and lung cancer within either group. The scenario described above is an example of which of the following?

- ☐ A) Observer bias
- ☐ B) Confounding
- ☐ C) Placebo effect
- ☐ D) Selective survival
- ☐ E) Nonresponse bias

Question: 2 of 4 [Qid : 51]

A cohort study is conducted to assess the relationship between oral contraceptive use and breast cancer. The study shows that in women with a family history of breast cancer, oral contraceptive use increases the risk of breast cancer with a relative risk (RR) of 2.10 and p value of 0.04. In women without a family history, no effect is observed (RR = 1.05, p = 0.40). The phenomenon described is an example of which of the following:

- ☐ A) Confounding
- ☐ B) Selection bias
- ☐ C) Latent period
- ☐ D) Effect modification
- ☐ E) Selective survival

Question: 3 of 4 [Qid : 52]

A case-control study is conducted to evaluate the association between alcohol consumption and cancer of the oral cavity. The crude analysis shows a strong association between the exposure and outcome: odds ratio = 4.5, 95% confidence interval 3.4 - 5.6. Smoking is considered as a potential confounder of the association. Which of the following properties of smoking is essential in order for it to be considered as a confounder?

- ☐ A) It must not be related to cancer of the oral cavity
- ☐ B) It must be prevalent in the population of interest
- ☐ C) It must be related to alcohol consumption
- ☐ D) It must be observed only in alcohol consumers
- ☐ E) It must not be controlled for in the analysis

Question: 4 of 4 [Qid : 53]

A case-control study is conducted to assess the relationship between alcohol consumption and breast cancer. First, the investigators interview patients with breast cancer. They then select neighbors of the patients with the same age and race to serve as controls. Such a study design helps to minimize which of the following problems?

- ☐ A) Selection bias
- ☐ B) Recall bias
- ☐ C) Observer's bias
- ☐ D) Effect modification
- ☐ E) Confounding

Correct Answers: 1) B 2) D 3) C 4) E

Explanation :

Confounding refers to the bias that results when the exposure-disease relationship of interest is mixed with the effect of extraneous factors (i.e., confounders). In order to be a confounder, the extraneous factor must have some properties linking it with the exposure and outcome of interest. An example of confounding bias is given in Question #1. Imagine that the results of the study described in Question #1 follow the pattern below:

	Alcohol Consumption		
Lung cancer	Yes	No	Total
Cases	60	40	100
Controls	40	60	100
Total	100	100	200

According to the results presented in the above table there is a strong association between alcohol consumption and lung cancer: odds ratio (OR) = $(60 \times 60) / (40 \times 40) = 2.25$. Once the investigators split the study subjects into smokers and non-smokers, however, the following results are obtained.

	Smokers Alcohol Consumption				Non-smokers Alcohol Consumption		
Lung cancer	Yes	No	Total	Lung cancer	Yes	No	Total
Cases	50	10	60	Cases	7	33	40
Controls	33	7	40	Controls	10	50	60
Total	83	17	100	Total	17	83	100

If you calculate the OR from each table the result in each case is 1.06. That means that there is no association between alcohol consumption and lung cancer once smoking status is accounted for. The statistical method of group separation described above is called stratified analysis. The association between alcohol consumption and lung cancer disappears after accounting for smoking status because smoking status is a confounder. To be a potential confounder, the risk factor must be related both to the exposure and to the

outcome (see Question #3). You can see from the tables above that smoking is more common among cases (60 vs 40) and among alcohol consumers (83 vs 17). Therefore, the effect of alcohol consumption observed during the crude analysis is in fact attributable to confounding.

There are several ways to limit confounding in both the design and analysis stages of a study.

Design stage: Randomization is an effective tool used in clinical trials for control of both known and unknown confounders (see section 15 for clinical trials). Matching is another tool used to limit confounding and is commonly employed in case-control studies. Investigators identify potential confounding variables, and select controls with variables that match those of the cases. For example, in Question #4 age and race are identified as potential confounders. The control group is selected in such a manner that both groups (cases and controls) have similar distribution of age and race. Furthermore, cases and controls are chosen from the same neighborhood. Selecting neighbors as controls has another advantage: it matches the cases to controls by variables that are difficult to measure (e.g., socioeconomic status). Restriction refers to limiting study inclusion by setting certain criteria (e.g., age, severity of the disease). The downside of restriction is that it limits generalizability (or external validity) of the study results.

Analysis stage: During analysis, confounding can be dealt with through stratified analysis as described above. More complicated statistical modeling methods are also commonly used to isolate the effect of exposure from the effects of various confounding factors.

Effect modification occurs when the effect of the exposure of interest on outcome is modified by another variable. In Question #2, the effect of oral contraceptive use on the incidence of breast cancer is modified by the family history: women with a positive family history have an increased risk, while women without a positive family history do not have an increased risk. Other well-known examples of effect modification include: 1) the effect of estrogens on the risk of venous thrombosis (modified by smoking), and 2) the risk of lung cancer in people exposed to asbestos (modified by smoking). Effect modification is NOT a bias. It is not due to flaws in either the design or analysis phase of the study. Effect modification is a natural phenomenon that should be described in the study's discussion section, but which cannot be corrected or eliminated.

Clinical Trials

Question: 1 of 4 [Qid : 54]

A clinical study is conducted to assess the role of non-specific beta-blockers in secondary prevention of variceal bleeding. Patients with liver cirrhosis surviving the first episode of variceal bleeding are treated with propranolol. The drug assignment (propranolol vs. placebo) is performed randomly. After patients have agreed to participate in the study, a computer assigns a random number to each patient which places him or her in one of the two groups. This drug assignment strategy is most helpful for controlling which of the following?

- ☐ A) Placebo effect
- ☐ B) Recall bias
- ☐ C) Selective survival
- ☐ D) Effect modification (interaction)
- ☐ E) Confounding

Question: 2 of 4 [Qid : 55]

A clinical trial is designed to evaluate the effect of a beta-blocker on the survival of patients with class IV heart failure. The beta-blocker or placebo therapy is given to patients along with standard therapy for heart failure. Neither the patient nor clinicians are aware of the drug (beta-blocker or placebo) that the patient is taking. The latter study design feature is used to prevent which of the following?

- ☐ A) Placebo effect and nonresponse bias
- ☐ B) Placebo effect and observer bias
- ☐ C) Recall bias and confounding
- ☐ D) Confounding and defaulting
- ☐ E) Lead-time bias and non-compliance

Question: 3 of 4 [Qid : 56]

A large-scale double-blind randomized clinical trial is conducted to assess the effect of a new aldosterone antagonist on the mortality and morbidity of congestive heart failure, class III-IV. 2,000 patients are enrolled: 1200 are assigned to the drug and 800 are assigned to placebo. According to the study results, patients treated with the new drug have improved survival ($RR = 0.85$, $p = 0.02$) and decreased risk of hospitalization ($RR = 0.65$, $p < 0.01$). The investigators also report that 10% of the placebo group and 14% of the treatment group discontinued therapy and that an additional 6% of patients in the placebo group were prescribed a different aldosterone antagonist. It is described in the statistical methods that the analysis was performed using the 'intention-to-treat' approach. Which of following is the best statement concerning the benefits of 'intention-to-treat'?

- ☐ A) Decreases placebo effect
- ☐ B) Decreases observer's bias
- ☐ C) Preserves the advantages of randomization
- ☐ D) Measures the degree of non-compliance
- ☐ E) Increases the power of the study

Question: 4 of 4 [Qid : 57]

A large-scale clinical trial is conducted to evaluate the effect of the beta-blocker therapy on the survival of patients with chronic heart failure, class IV. The patients with severe heart failure are randomly assigned to carvedilol, a beta-blocker or to placebo. In their report of the study results, the investigators include a table with baseline characteristics (age, race, prevalence of hypertension, etc) of the patients in the treatment and placebo groups. According to the table, both groups have similar distributions of these characteristics. The similar distributions of these characteristics best reflects which of the following:

- ☐ A) Sample size is adequate
- ☐ B) The study is negative
- ☐ C) The power of the study is high
- ☐ D) Randomization is successful
- ☐ E) Observer's bias might be an issue

Correct Answers: 1) E 2) B 3) C 4) D

Explanation :

Randomized clinical trials are a type of interventional (experimental) study design (see Section 12) and can provide the strongest evidence regarding an exposure-disease relationship. Several important features of randomized clinical trials are discussed below. These are randomization, blinding and 'intention-to-treat' analysis.

Randomization implies exposure assignment that is determined by chance. Neither the investigator nor the study subject has any control over placement. The goal of randomization is to create groups with similar distributions of known (as described in Question #4) and unknown variables, the only difference being the exposure assigned. Randomization therefore minimizes the effect of confounding (see section 14). It also eliminates the possibility of susceptibility bias, whereby the care provider systematically assigns patients to specific groups based in part on the severity of disease (see section 13).

Blinding refers to the study design technique whereby exposure status is kept hidden from the patient and/or the investigator. In single-blinded studies, patients are not aware whether they are taking the drug or placebo. This minimizes the placebo effect. The placebo effect can be especially significant in studies measuring subjective symptoms (e.g., frequency of headaches, or overall wellbeing). In double-blinded studies, both the patient and caregiver are unaware of the exposure status of the patient. Blinding the caregiver prevents conscious or unconscious misclassification of outcomes by the caregiver, a phenomenon called observer bias.

Intention-to-treat is an important principle used in the analysis of randomized clinical trials. Intention-to-treat means that the patient's treatment status at the point of randomization is analyzed. If a patient who is assigned to the placebo group begins taking the medication assigned to the treatment group sometime after study initiation, or if a patient in the treatment group stops taking the prescribed medication, the data from these patients is still analyzed along with their original group. The value in the intention-to-treat approach is that it preserves the benefits of randomization and prevents bias due to selective non-compliance. Investigators may alternatively use the 'as treated' rule, which is the opposite of intention-to-treat (i.e. if a patient switches therapy they are counted as members of the new group during analysis).

Statistical Distributions

Question: 1 of 4 [Qid : 58]

A study of 400 patients hospitalized with diabetes mellitus-related complications shows that serum cholesterol level is a normally distributed variable with mean of 230 g/dl and standard deviation of 10 mg/dl. Based on the study results, how many patients do you expect to have serum cholesterol ≥ 250 mg/dl in this study?

- ☐ A) 2
- ☐ B) 10
- ☐ C) 20
- ☐ D) 64
- ☐ E) 128

Question: 2 of 4 [Qid : 59]

A large study of serum cholesterol levels in patients with diabetes mellitus reveals that the parameter is normally distributed with a mean of 230 mg/dL and standard deviation of 10 mg/dL. According to the results of the study, 95% of serum cholesterol observations in these patients lie between which of the following limits?

- ☐ A) 220 and 240 mg/dL
- ☐ B) 225 and 235 mg/dL
- ☐ C) 210 and 250 mg/dL
- ☐ D) 200 and 260 mg/dL
- ☐ E) 220 and 260 mg/dL

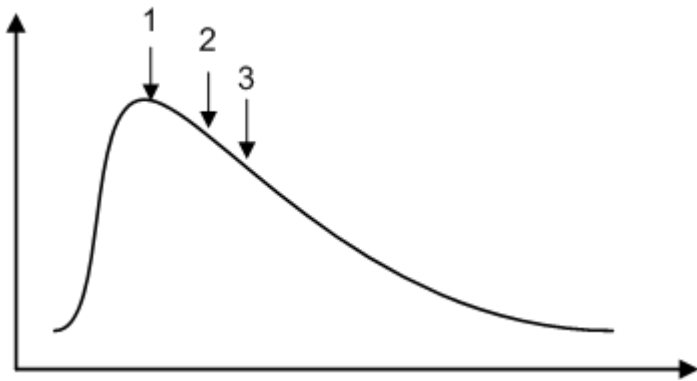
Question: 3 of 4 [Qid : 60]

A patient has his blood glucose level measured. The population mean blood glucose level is then subtracted from the patient's blood glucose level. The result is then divided by the standard deviation. If we assume that the blood glucose level in the population follows a normal distribution, the value obtained is best referred to as:

- ☐ A) T score
- ☐ B) Z score
- ☐ C) F value
- ☐ D) Chi-square value
- ☐ E) Correlation coefficient

Question: 4 of 4 [Qid : 61]

HbA1c level is measured in diabetic patients placed on an intensive insulin therapy. The distribution of the values is shown on the slide below.



Which of the values indicated on the slide most likely correspond to the mean, median and mode, respectively?

- ☐ A) 3, 2, 1
- ☐ B) 3, 1, 2
- ☐ C) 2, 3, 1
- ☐ D) 2, 1, 3
- ☐ E) 1, 2, 3
- ☐ F) 1, 3, 2

Correct Answers: 1) B 2) C 3) B 4) A

Explanation :

Normal distribution is the most common statistical distribution tested on USMLE exams. Many real-life continuous parameters follow normal distribution (e.g. systolic blood pressure, serum potassium level, blood glucose level, etc.). There are several properties that help to define normal distribution:

- Graphically, a normal distribution forms a symmetric bell-shaped curve.
- The mean, median and mode of a variable that follows normal distribution are equal or very close to each other.
- The 68/95/99 rule holds for normal distribution. It states that 68% of all observations lie within 1 standard deviation of the mean, 95% lie within 2 standard deviations, and 99.7 % lie within 3 standard deviations.

In Question #1, the cutoff point of 250 mg/dl is 2 standard deviations above the mean, leaving a tail of 2.5% to the right (2.5% of 400 patients equals 10 patients). Fig. 7 demonstrates the point.

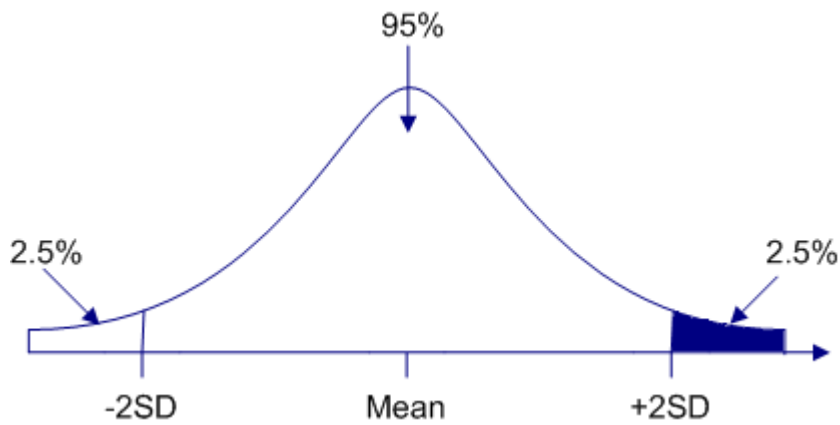


Fig. 7: 95% of observations in normal distribution lie within 2 standard deviations of the mean, leaving 2.5% of observation at each tail.

A normal distribution with the mean of 0 and variance of 1 is called a standard normal distribution. Any variable that follows a normal distribution can be transformed to a standard normal distribution by using the approach described in Question #3 (subtracting the mean from all values and then dividing by the standard deviation). When this process is applied to any given value in the data set, the value's Z-score is obtained. The Z score indicates how many standard deviations a given value is from the mean.

Skewed distributions are asymmetric, having a tail either to the right (positively skewed) or to the left (negatively skewed). A typical positively skewed distribution is shown in Question #4. Mode of a positively skewed distribution corresponds to the peak of the curve. Median is further to the right because it bisects the number of observations whereas mean is even further to the right because it is affected by high values at the right tail.

Comparing Groups

Question: 1 of 4 [Qid : 62]

An investigator compares an average standardized depression score in two groups of hypertensive patients: those who take beta-blockers and those who do not. Which of the following tests is most likely to be employed by the investigator to analyze the study results?

- ☐ A) Paired t test
- ☐ B) Two-sample t test
- ☐ C) Fisher's exact test
- ☐ D) Pearson's chi-square test
- ☐ E) Analysis of variance
- ☐ F) Spearman's correlation coefficient

Question: 2 of 4 [Qid : 63]

A study is conducted to assess the association between hormone replacement therapy (HRT) in post-menopausal women and the level of serum C-reactive protein (CRP). The data from the study are presented below:

	CRP high	CRP normal	
HRT	32	41	73
No HRT	28	49	77
	60	90	150

Which of the following is the best statistical method to assess the association between HRT and elevated CRP levels?

- ☐ A) Paired t test
- ☐ B) Two-sample t test
- ☐ C) Fisher's exact test
- ☐ D) Pearson's chi-square test
- ☐ E) Analysis of variance
- ☐ F) Spearman's correlation coefficient

Question: 3 of 4 [Qid : 64]

It is claimed that a new drug induces rapid and sustained weight loss by affecting triglyceride metabolism in the small intestine. The body mass index of 100 patients is calculated at baseline and compared to the value after 1 year of treatment with the drug. Which of the following tests is most likely to be employed by the investigators to analyze the study results?

- ☐ A) Paired t test
- ☐ B) Two-sample t test
- ☐ C) Fisher's exact test

- ☐ D) Pearson's chi-square test
- ☐ E) Analysis of variance
- ☐ F) Spearman's correlation coefficient

Question: 4 of 4 [Qid : 65]

A clinical study evaluates the role of thymectomy in patients with myasthenia gravis who do not have an anterior mediastinal mass on chest CT scan. Out of 9 patients who undergo thymectomy, 7 show sustained improvement after one year of follow-up. Out of 20 patients treated conservatively, 8 show sustained improvement after one year of follow-up. Which of the following tests is most likely to be employed by the investigators to analyze the study results?

- ☐ A) Paired t test
- ☐ B) Two-sample t test
- ☐ C) Fisher's exact test
- ☐ D) Pearson's chi-square test
- ☐ E) Analysis of variance
- ☐ F) Spearman's correlation coefficient

Correct Answers: 1) B 2) D 3) A 4) C

Explanation :

The algorithm presented in Fig.8 helps identify the correct statistical test to apply in common situations:

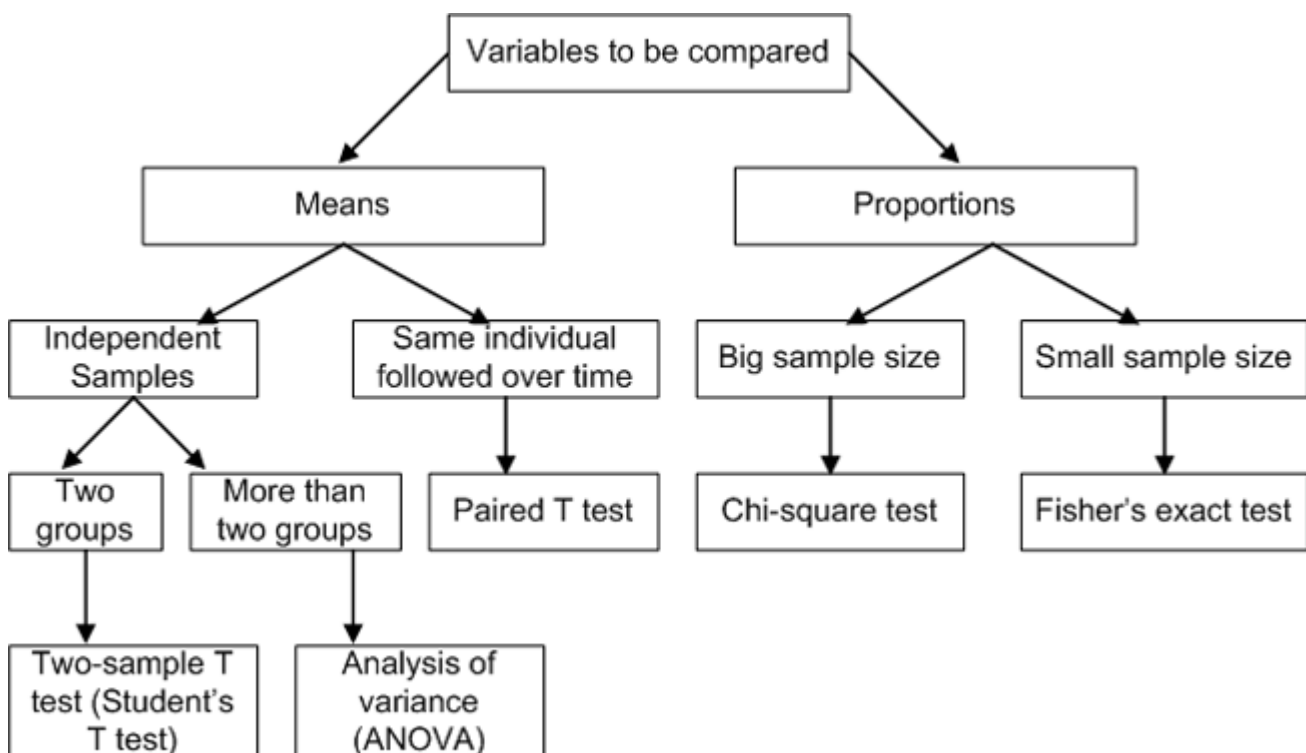


Fig. 8. The algorithm helps identify the correct statistical test in common situations.

The Two-sample t test (also called Student's t test) is commonly employed to compare means of two independent groups. The basic requirements needed to perform this test are the two mean values, the sample variances, and the sample size. The t statistic is then obtained to calculate the p value. If the p value is less than 0.05, the null hypothesis (that there is no difference between the two groups) is rejected, and the two means are assumed to be statistically different. If the p value is large, the null hypothesis is retained.

The Paired t test is also used to compare two means but unlike the Student's t test it is used in situations where the means are dependent. A typical situation is described in Question #3: two means from the same individual (baseline BMI and BMI after treatment) are compared.

Analysis of variance (ANOVA) is used to compare means of three or more variables.

The Chi-square test is used to compare the proportions of a categorized outcome. In Question #2, outcome (serum CRP level) is categorized as either "high" or "normal," and then presented with exposure ("HRT" or "no HRT") in a 2 x 2 contingency table. In a typical Chi-square test, the observed values in each of the cells are compared to expected (under the hypothesis of no association) values. If the difference between the observed and expected values is large, an association between the exposure and the outcome is assumed to be present. The Chi-square test can be employed for a large sample size. If the sample size is small, Fisher's exact test is used. It is typically preferred for situations when an expected value in either of the cells is less than 10. In Question #4, a study with a small sample size is described and Fisher's exact test would be the best way to analyze the results.

Survival Analysis

Question: 1 of 3 [Qid : 66]

A study of patients with pancreatic cancer assesses the efficacy of a new chemotherapy regimen. The table below presents survival information for patients treated with the new regimen:

Time, in months	Number of patients at the beginning of the interval	Number of patients who died during the interval	Percentage of patients who died during the interval
0-1	200	20	10
1-2	180	10	5.6
2-3	170	12	7
3-4	158	18	11
4-5	140	20	14

What is the probability that a patient on the new regimen is alive at 3 months?

- ☐ A) 0.93
- ☐ B) 0.89
- ☐ C) $(0.9 + 0.94 + 0.93)/3$
- ☐ D) $0.9 \times 0.94 \times 0.93$
- ☐ E) $1 - 0.89 \times 0.86$

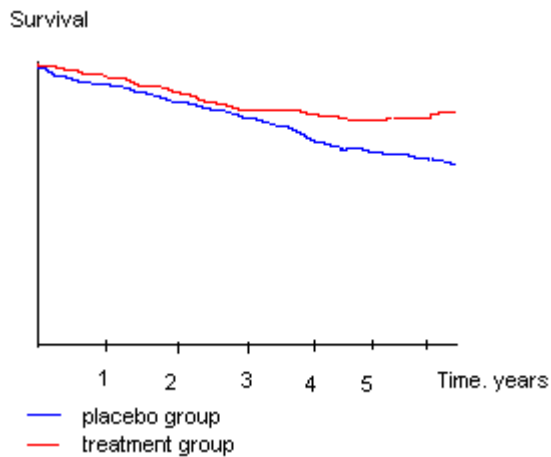
Question: 2 of 3 [Qid : 67]

A randomized double-blinded clinical trial is conducted to assess the role of multidrug chemotherapy in the treatment of patients with stage III – IV stomach cancer. 150 patients in the treatment group and 100 patients in the placebo group are followed for 24 months. 120 patients in the treatment group (80%) and 80 patients in the placebo group (80%) die during the follow-up period. The investigators conclude that the treatment is effective. Which of the following is the most likely explanation for such a conclusion?

- ☐ A) Observer bias may be present
- ☐ B) Selective survival may be an issue
- ☐ C) The results are confounded
- ☐ D) Time-to-event data were analyzed
- ☐ E) Two-year risk was calculated

Question: 3 of 3 [Qid : 68]

A large-scale clinical trial is conducted to assess the effect of a multi-vitamin supplement on the risk of future cardiovascular events. The outcomes measured by the study are cardiovascular mortality, non-fatal myocardial infarction and coronary revascularization procedures. According to the study results, the overall relative risk of the cardiovascular outcomes for the placebo group compared to the treatment group was 1.5, $p = 0.30$, although the relative risk for the 5th year of follow-up was 2.05, $p = 0.01$. Survival curves for the two groups were parallel during the first 3 years of observation, but began to separate the 3rd year, favoring the treatment group.



Which of the following statements is true concerning the study results given above?

- ☐ A) Multi-vitamin use seems to be ineffective in preventing cardiovascular events
- ☐ B) Inappropriate selection of the study subjects may be present
- ☐ C) Latent period can be demonstrated on the survival plot
- ☐ D) The follow-up period is too long for such a study
- ☐ E) The sample size is not large enough and the measure of outcome is unstable

Correct Answers: 1) D 2) D 3) C

Explanation :

Time-to-event data analysis is becoming more and more popular for analyzing follow-up studies and clinical trials. This type of analysis is called '*survival analysis*'. A simple data layout for survival analysis is shown in Question #1. Rows are arranged by time intervals. In each row, data on the number of subjects who were present at the beginning of the time interval and the number who died during the interval are provided. Therefore probabilities of mortality/survival can be calculated for each time interval. For example, the probability for a patient to survive one additional month once he/she already survived the first two months of chemotherapy would be 93%. Cumulative probability can be calculated by multiplying individual probabilities. For example, the probability that a patient on the new regimen would survive at least 3 months is the product of three probabilities ($0.9 \times 0.94 \times 0.93$).

It is important to understand that survival analysis accounts not only for the number of events in both groups, but also for the timing of the events. Despite the fact that two-year mortality risk is the same for both groups in Question #2, the patients in the treatment group may on average live longer than the patients in the placebo group. For example, the median survival time may be 3 months for the placebo group and 9 months for the treatment group. Therefore, in Question #2 time-to-event analysis could explain the conclusion that treatment was effective despite equal mortality at two years..

A survival plot represents a graphical description of survival analysis. An example is shown in Question #3. The concept of a latent period is demonstrated in this case. Latency is a very important issue to consider in chronic disease epidemiology. The latent period between exposure and the development of an outcome is relatively short in infectious diseases. In chronic diseases (e.g., cancer or coronary artery disease), however, there may be a very long latency period. In Question #3, at least three years of continuous exposure to multivitamins are required to reveal the protective effect of the exposure on

cardiovascular outcomes. On the survival plot, you can clearly see that the survival curves run parallel to each other for three years (the latent period), and then begin to separate at the 3rd year of follow-up. Overall relative risk is not statistically significant, because it is 'diluted' by the years of latency, although the relative risk for the 5th year of follow-up, when isolated, clearly demonstrates the beneficial effect of therapy.

Statistical Power

Question: 1 of 3 [Qid : 69]

A randomized double-blind clinical trial is conducted to evaluate the effect of a new hypolipidemic drug on the survival of patients after PTCA. 1000 patients undergoing PTCA are randomly assigned to the drug or placebo (500 patients in each group) and then followed for 3 years for the development of acute coronary syndrome. Severe acute myositis is reported as a rare side effect of the drug therapy, but the difference between the two groups in the occurrence of this side effect is not statistically significant ($p = 0.09$). The same side effect was reported in several small clinical trials of this drug. The failure to detect a statistically significant difference in the occurrence of acute myositis between the treatment and placebo groups is most likely due to:

- ☐ A) Selection bias
- ☐ B) Short follow-up period
- ☐ C) Inappropriate selection of the patients
- ☐ D) Small sample size
- ☐ E) Observer's bias

Question: 2 of 3 [Qid : 70]

The researchers want to further investigate the association between the new hypolipidemic drug and the occurrence of severe acute myositis. They note that several other studies have reported this side effect, but none of these studies demonstrated a statistically significant difference in rates of severe acute myositis between the treatment and placebo groups. The best method to further investigate a possible association between the drug and development of severe acute myositis is to:

- ☐ A) Conduct a new large-scale clinical trial
- ☐ B) Review the medical charts to re-ascertain the events
- ☐ C) Do stratified analysis on multiple risk-factors
- ☐ D) Pool the data from several trials
- ☐ E) Ignore the possible association between the drug and acute myositis

Question: 3 of 3 [Qid : 71]

A large prospective study is designed to assess the association between postmenopausal hormone replacement therapy (HRT) and the risk of dementia, Alzheimer type. Small studies conducted earlier suggest a possible protective effect of HRT. What is the probability that the study will show an association if in fact HRT does affect the risk of dementia?

- ☐ A) α
- ☐ B) β
- ☐ C) $1 - \alpha$
- ☐ D) $1 - \beta$
- ☐ E) Type I error
- ☐ F) Type II error

Correct Answers: 1) D 2) D 3) D

Explanation :

With any scientific study, there is always the risk of reaching an incorrect conclusion. Incorrect conclusions come in two main forms:

1) Wrongfully concluding that there is an association between exposure and disease when in fact there is none. Such error is called type I error.

2) Wrongfully concluding that there is no association between exposure and outcome, when in fact there is one. Such error is called type II error.

The probability of committing type I error is referred to as alpha and is expressed in epidemiological and clinical studies as the p value. For example, a p value of 0.04 means there is still a 4% chance that no association exists between exposure and outcome even though the null hypothesis has been rejected. In most studies, the alpha level (also called the statistical significance level) is set to 0.05; that means researchers can reject the null hypothesis only if its probability of being true is less than 5%.

The probability of committing type II error is referred to as beta. $(1 - \beta)$ indicates the probability of detecting an association if it exists in reality and is referred to as the "power of the study".

The power of a study depends on the following factors:

- Alpha level (statistical significance level): Lowering the alpha level (i.e., strengthening the significance criterion) decreases the power of the study.
- The magnitude of difference in outcome between the study groups (i.e. a subtle difference is more difficult to detect than a big difference).
- Increasing the sample size increases the probability of detecting a difference in outcome between the study groups.

As described in Question #1, while acute myositis was reported in several clinical trials of the drug, in this study the result was not statistically significant. Because this side effect is rare and few patients experienced it, the limited size of the study group resulted in a p value that did not reach statistical significance. A bigger sample size would increase the ability to detect the difference (i.e., power of the study) and likely result in a lower, statistically significant p value. Increasing the follow-up period would not increase the incidence of the severe acute myositis if this side effect occurs in susceptible individuals during only the early stages of therapy. Therefore, increasing the sample size would be the best approach.

Pooling together for analysis the data from several studies is called meta-analysis. Meta-analysis is a useful epidemiologic tool that is employed to increase the power of the data. If the outcome is rare or the difference between the groups is small it may be difficult for a single study (even one that is large-scale) to detect the difference and reach statistical significance. In that case meta-analysis can be used to increase the sample size and therefore the power of the analysis. The major disadvantage of meta-analysis is that while it pools together the data from many studies, it also 'pools' together the biases and limitations of those individual studies.

Variability and Validity

Question: 1 of 2 [Qid : 72]

An HIV-positive patient with a two-day history of fever is seen by three doctors in the hospital. Two of the doctors record crackles in the left lung base and diagnose community-acquired pneumonia. The third doctor reports clear lungs. Which of the following phrases best describes the role of auscultation as a diagnostic tool in this case?

- ☐ A) Not valid
- ☐ B) Not reliable
- ☐ C) Not sensitive
- ☐ D) Not specific
- ☐ E) Not accurate

Question: 2 of 2 [Qid : 73]

A case-control study is conducted to assess the role of occupational exposure to certain chemicals in the development of pancreatic cancer. The study fails to demonstrate an association between documented exposures and pancreatic cancer. Which of the following does not affect validity of the study?

- ☐ A) Selection bias
- ☐ B) Differential misclassification
- ☐ C) Confounding
- ☐ D) Sample size

Correct Answers: 1) B 2) D

Explanation :

Results of any epidemiological or clinical study as well as any diagnostic test can be affected by two broad categories of error: random error and systematic error.

Random error is explained by chance and therefore is unpredictable. The terms that describe the degree of random variation include precision. Precision addresses the scope of random variation in study results and can be quantified as the reciprocal of variance. It also refers to reliability, or reproducibility of measurements. Inter-rater reliability describes the degree of similarity in test results obtained by different investigators. A lack of inter-rater reliability is demonstrated in Question #1.

Systematic error or bias is caused by flaws in study design and/or analysis and is not a product of chance. Unlike random error, if a second investigator were to perform the same study or diagnostic test under the same conditions, he or she would reliably achieve the same (systematic) error. Systematic error compromises the validity of the study. In contrast to random error, systematic error is not affected by sample size. Forms of systematic error are covered in other sections (selection and misclassification bias are covered in section 13; confounding is covered in section 14).